Taxing Top Incomes in a World of Entrepreneurs

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Abstract

This paper studies the aggregate effects of top marginal income taxation in an economy where entrepreneurs accumulate firm productivity over time and choose when to incorporate. I identify two channels through which higher top income tax rates distort entrepreneurial decisions. The first is the productivity accumulation effect. For many entrepreneurs, reaching the top income bracket is the result of operating and growing the same firm for many years. A higher top tax rate reduces the future payoff to firm growth, inducing them to slow productivity accumulation even before they become top income earners. The second is the incorporation timing effect. A small group of highly able entrepreneurs reach the top income bracket soon after starting a business, so slowing firm growth offers little scope to avoid the higher tax rate. Instead, they accelerate productivity accumulation and incorporate earlier, transferring younger firms with less accumulated productivity to the corporate sector. Together, these responses reduce firms' accumulated productivity and lower aggregate output.

Keywords: Macroeconomics, Entrepreneurship, Firm dynamics, Income and Wealth distributions, Taxation

JEL Codes: H21, H24, H25, L26, D31.

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1. Introduction

The taxation of top incomes has long been at the center of policy debates. Higher top marginal income tax rates may reduce inequality, but they may also distort economic decisions and reduce aggregate economic activity. The magnitude of these effects depends in part on who earns top incomes and how they respond to taxation. In the United States, a substantial share of top income earners are entrepreneurs. Their responses are especially important because the decisions they make affect not only their own incomes but also the businesses they operate. This has motivated a growing quantitative literature studying the effects of top income taxation in economies with entrepreneurship (see, for example, Bruggemann, 2021).

Existing studies have examined how taxes affect occupational choice and entrepreneurs' saving, investment, and hiring decisions. These decisions are important for determining who becomes an entrepreneur and how much their firms expand. Yet growing a business involves more than expanding its capital and workforce. As entrepreneurs operate their firms, they can also improve production methods, learn to organize their businesses more efficiently, and build firm-specific knowledge. Over time, these improvements can make their firms more productive and shape how they grow. This raises another question: how does top income taxation affect the way entrepreneurs build up productivity within their firms over time? Even if higher tax rates do not change who becomes an entrepreneur, they may change the kinds of firms those entrepreneurs build and the growth paths those firms follow.

Moreover, the effects of top income taxation on firm growth may also extend to the corporate sector. This is particularly important because C corporations account for more than half of employment and nearly 60% of business output in the United States. However, existing quantitative models of entrepreneurship typically treat the corporate sector as a separate production sector, often represented by a single firm. As a result, top income taxation affects the corporate sector only through general equilibrium adjustments. This leaves out a potentially important connection: if higher top income taxes change how entrepreneurs build productivity within their firms, these effects may also reach the corporate sector when the firms incorporate. Therefore, this paper studies two channels that have received limited attention in the previous literature: how does top income taxation shape the growth of entrepreneurial firms over their life cycles, and how might these effects extend to the corporate sector through incorporation?

To study these questions, I extend the standard heterogeneous-agent general equilibrium life-cycle model of entrepreneurship developed by Quadrini (2000) and Cagetti and De Nardi (2006). I introduce two new ingredients. First, entrepreneurs can build their firms' productivity over time by accumulating an intangible asset called organizational capital. Organizational capital accumulates through production experience, reflecting learning by doing, and entrepreneurs differ in how efficiently they build it. As a result, firms run by entrepreneurs with different abilities may follow different growth paths. Second, entrepreneurs can choose to incorporate their businesses. I model incorporation as the transfer of an entrepreneurial firm to the corporate sector, where the firm retains its accumulated organizational capital and continues to grow. Upon incorporation, the entrepreneur receives the market value of the firm and pays capital gains tax on the proceeds rather than personal income tax. After incorporation, entrepreneurs can start new businesses, allowing them to build and incorporate multiple firms over their lifetimes. Together, these two ingredi-

ents allow top income taxation to affect not only how entrepreneurs grow their firms, but also the composition of firms entering the corporate sector.

I calibrate the model to the U.S. economy using both household-level and firm-level data. On the household side, I target the population share of entrepreneurs, income and wealth concentration at the top, and the importance of entrepreneurs among top income earners. On the firm side, I use moments that capture firm dynamics, including firm exit and survival rates and the employment shares of young and mature firms. I also target the shares of firms and employment accounted for by C corporations. An important feature of this calibration is that it disciplines the model using both household and firm dynamics moments. While household moments help discipline entrepreneurial ability heterogeneity and the distribution of income and wealth, firm dynamics moments provide additional information about how firms accumulate productivity, grow, and survive over time.

The calibrated model closely matches these targets and also performs well in reproducing several important moments that are not directly targeted. In particular, the model captures the firm size distributions of both pass-through businesses and C corporations. Since I only target the overall share of C corporations and their employment share, matching the size distributions provides an additional check on the share of C corporations across different firm size categories. The model also reproduces the declining pattern of entrepreneurs' business tenure observed in the data. This is informative because in the model, how long an entrepreneur operates a business depends on both firm exit and incorporation decisions. Matching this pattern provides an additional check on the model's implied firm dynamics and incorporation behavior. These untargeted results lend further support to the model's ability to capture firm life-cycle dynamics and the allocation of business activity across the entrepreneurial and corporate sectors.

The calibrated economy features two distinct patterns of entrepreneurship, depending on entrepreneurial ability. Entrepreneurs with relatively low ability accumulate organizational capital slowly. Among those who eventually reach the top income bracket, most do so after operating the same firm for many years and gradually building up its productivity. Even after reaching the top income bracket, they tend to continue operating their businesses rather than incorporating, as their relatively low ability makes it difficult to build another successful firm. I refer to them as lifelong entrepreneurs. In contrast, high-ability entrepreneurs can accumulate organizational capital much more quickly, allowing them to reach the top income bracket soon after starting a business. They also tend to incorporate relatively early. Since their high ability allows them to build productive firms quickly, they often return to entrepreneurship by starting new businesses after incorporation, repeating this process over their working lives. I refer to them as serial entrepreneurs.

The calibrated economy features two distinct entrepreneurial career paths, shaped by differences in entrepreneurial ability. Most entrepreneurs have relatively low ability and accumulate organizational capital slowly as they operate their businesses. Among those who eventually reach the top income bracket, most do so after running the same firm for many years and gradually building up its productivity. Across the lower ability groups, the median time from starting a business to reaching the top income bracket is around 10 years, conditional on reaching it. These entrepreneurs are also less likely to incorporate their businesses. Given their limited ability to accumulate organizational capital, building another productive firm after incorporation would take many years. They therefore tend to continue operating their existing firms. I refer to this group as

lifelong entrepreneurs.

In contrast, entrepreneurs with high ability can accumulate organizational capital much faster. They reach the top income bracket soon after starting a business and typically incorporate shortly afterward. Their high ability also makes it easier to start over after incorporation, as they can rapidly build up productivity in a new firm and repeat the process. For entrepreneurs with the highest ability, for example, the median time to reach the top income bracket is only 3 years, and the median time to incorporation is 4 years. On average, they incorporate around 7 businesses over their lifetime. I refer to these entrepreneurs as *serial entrepreneurs*.

Although serial entrepreneurs are more likely to reach the top income bracket, they account for only a small fraction of all entrepreneurs. In the baseline economy, serial entrepreneurs account for less than 7% of all entrepreneurs but about 20% of those in the top income bracket. The remaining 80% belong to the lifelong group. Therefore, top income taxation affects entrepreneurs with very different business histories: some have spent many years building up the productivity of a single firm, while others reach the top bracket relatively early and repeatedly start and incorporate businesses. These differences in entrepreneurial career paths are important for understanding how entrepreneurs respond to changes in top income taxation.

I then use the calibrated model to study the effects of increasing the top marginal income tax rate. In this experiment, I hold all other tax parameters and government spending fixed at their benchmark levels and balance the government budget through a uniform lump-sum transfer. For each tax rate, I solve for a new general equilibrium steady state and compare it with the benchmark economy. Total tax revenue reaches its maximum at a top income tax rate of 58%, an increase of 21 percentage points from the benchmark rate. At this rate, total tax revenue is 5.5% higher than in the benchmark economy. While most of the revenue increase comes from personal income taxation, corporate income and capital gains tax revenues also rise. However, the higher tax rate comes with economic costs: aggregate output falls by 2.9%, and aggregate organizational capital declines by 4.9%.

The aggregate decline in organizational capital reflects two distinct responses to higher top income taxation. The first is the *productivity accumulation effect*, which primarily affects lifelong entrepreneurs. As discussed above, these entrepreneurs typically spend many years building productivity within the same firm before reaching the top income bracket. A higher top marginal income tax rate reduces the expected future payoff from firm growth, inducing them to slow productivity accumulation and delay entering the top income bracket. This forward-looking response discourages firm growth even before these entrepreneurs become subject to the higher tax rate.

The second is the *incorporation timing effect*, which primarily affects serial entrepreneurs. Their high ability allows them to accumulate productivity rapidly and reach the top income bracket soon after starting a business. Slowing productivity accumulation offers them limited scope to avoid the higher tax rate. Instead, a higher top marginal income tax rate makes incorporation more attractive, as they can realize the value of their firms as capital gains, which are taxed at a lower rate than ordinary income. Serial entrepreneurs respond by accelerating productivity accumulation so that they can incorporate earlier. Although productivity grows faster, the shorter time before incorporation makes these firms enter the corporate sector with less accumulated organizational

capital.

This earlier incorporation also changes the composition of firms in the entrepreneurial sector. After incorporation, serial entrepreneurs typically start new businesses and build up organizational capital again in their new firms. As their established firms leave the entrepreneurial sector sooner and new businesses take their place, the sector has fewer older, larger firms and a greater share of young, smaller ones. These changes, together with slower productivity accumulation by lifelong entrepreneurs, contribute to the decline in organizational capital in the entrepreneurial sector. At the revenue-maximizing tax rate, aggregate organizational capital in this sector falls by 7.3%.

Higher top income taxation also changes the composition of firms entering the corporate sector. In addition to earlier incorporation by serial entrepreneurs, higher taxes also lead some lifelong entrepreneurs to incorporate. Although these entrepreneurs slow productivity accumulation to delay entering the top income bracket, some eventually reach it. Once they do, the higher tax burden on their business profits makes incorporation more attractive than continuing to operate their firms, despite their limited ability to build another successful business. As a result, some incorporate firms that they would have continued operating in the benchmark economy.

Serial entrepreneurs incorporate their businesses at younger ages, while the additional firms incorporated by lifelong entrepreneurs tend to have less organizational capital than those incorporated by serial entrepreneurs. Overall, these changes bring more firms with relatively little accumulated organizational capital into the corporate sector. As a result, the number of corporate firms increases by 6.5%, while average organizational capital per corporate firm declines by 2.2%. Despite the decline in average organizational capital, the larger number of corporate firms raises aggregate organizational capital in this sector by 4.1%. However, this gain is insufficient to offset the decline in the entrepreneurial sector, resulting in a 4.9% decline in aggregate organizational capital for the whole economy.

Finally, I evaluate the welfare implications of higher top income taxation using the consumption-equivalent variation of a newborn under the veil of ignorance. The welfare-maximizing top marginal income tax rate is 41%, only 4 percentage points above the benchmark rate and well below the revenue-maximizing rate of 58%.[1] In my model, further increases in the top tax rate generate diminishing revenue gains, while the welfare costs associated with distorted entrepreneurial decisions and lower aggregate production continue to rise. As a result, the additional revenue used to finance lump-sum transfers is no longer sufficient to offset these welfare costs beyond a top marginal income tax rate of 41%.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model, followed by the calibration strategy in Section 4 and the key features of the baseline economy in Section 5. Section 6 describes the policy experiment and presents the results. Section 7 concludes.

[1]For comparison, Bruggemann (2021) reports a welfare-maximizing top marginal income tax rate of 60% in a standard quantitative model of entrepreneurship without endogenous firm productivity accumulation or incorporation decisions.

2. Related Literature

This paper relates to four strands of literature. The first studies top income taxation in quantitative macroeconomic models, with a growing focus on the role of entrepreneurship. The second examines how taxation affects endogenous productivity growth and the accumulation of intangible capital. The third studies taxation and entrepreneurs' choice of legal form of organization. The fourth is the firm dynamics literature. The contribution of this paper is to connect these margins in a framework where top income taxation affects how entrepreneurs build their firms and when they incorporate, with consequences for both the entrepreneurial and corporate sectors.

This paper first contributes to the quantitative literature on top income taxation.[2] A large literature studies the effects of top marginal income tax rates in dynamic heterogeneous agent models without entrepreneurship, including Guner et al. (2016), Badel et al. (2020), and Kindermann and Krueger (2022). Motivated by the importance of entrepreneurs at the top of the income and wealth distributions (Gentry and Hubbard, 2004; Buera et al., 2015), more recent work has introduced entrepreneurship into this discussion. Examples include Bruggemann (2021), İmrohoroğlu et al. (2023), Holter et al. (2023), and Ge (2026). These papers build on the standard quantitative entrepreneurship framework of Quadrini (2000) and Cagetti and De Nardi (2006), in which households choose between wage work and entrepreneurship and firm productivity is exogenous. The entrepreneurial and corporate sectors are also separate, so firms do not move between them through incorporation. My model retains the standard occupational choice, saving, and investment margins, but adds endogenous productivity accumulation and an incorporation decision that links the two sectors. These additions allow top income taxation to affect not only whether households become entrepreneurs and how much they invest, but also how firms grow over their life cycles and when entrepreneurial businesses are transferred to the corporate sector.

More broadly, the taxation literature has studied endogenous productivity accumulation and the choice of legal form as two important margins of adjustment, but these margins have typically been analyzed separately. A second strand of related work allows taxation to affect productivity through endogenous investment in ideas or intangible assets. Jones (2022) studies top income taxation in an economy where successful innovations generate top incomes and new ideas drive aggregate growth. Higher top marginal tax rates discourage innovation and slow the creation of new ideas, so their effects extend beyond the incomes of those directly subject to the tax. Bhandari and McGrattan (2021) develop a model in which private business owners build sweat equity by investing their time and resources in customer bases, client lists, and other intangible assets. They show that allowing for this endogenous intangible asset accumulation substantially increases the response of private business activity to business income taxes relative to models that abstract from sweat equity. My paper shares with these studies the idea that taxation can affect aggregate outcomes by changing incentives to build productive assets. The key difference is that productivity accumulation in my model takes place within entrepreneurial firms over their life-cycles and interacts with entrepreneurs' incorporation decisions. This interaction allows top income taxation to affect not only productivity accumulation in the entrepreneurial sector, but also outcomes in the corporate sector through incorporation.

[2]Earlier work also studies optimal top income taxation in static settings. See, for example, Diamond and Saez (2011) and Piketty and Saez (2013).

The paper is also related to a large literature on taxation and firms' choice of legal form of organization (see, for example, Chen et al., 2018; Bilicka and Raei, 2023). Among this literature, my paper is most closely related to Dyrda and Pugsley (2022) and Di Nola et al. (2025). Dyrda and Pugsley (2022) develop a quantitative general equilibrium model in which business owners choose between pass-through entities and C corporations, and use the model to study how tax policy affects the allocation of business activity across legal forms. Di Nola et al. (2025) study top income taxation in a model where entrepreneurs choose their legal form of organization and can shift income across tax bases. They find that higher top marginal income tax rates induce high-income entrepreneurs to switch toward C corporations. My model has a similar prediction that higher personal income taxation increases incorporation. The key difference is that firms in my model accumulate productivity over their life-cycles. Higher top income taxation changes both entrepreneurs' productivity accumulation and the timing of incorporation. Together, these responses affect the productivity of firms entering the corporate sector. As a result, top income taxation affects not only entrepreneurial firm growth and incorporation decisions, but also the composition of firms in the corporate sector. This creates a channel through which distortions to entrepreneurial firm growth are transmitted across sectors and affect aggregate productivity and resource allocation.

The paper contributes to the firm dynamics literature, which studies how heterogeneity in firm productivity and growth shapes aggregate outcomes (Hopenhayn, 1992; Haltiwanger, 2012; Haltiwanger et al., 2013). More recent work emphasizes that differences in firms' growth potential at entry can have persistent effects on firm performance and aggregate dynamics (Sedlacek and Sterk, 2017; Sterk et al., 2021). Related empirical evidence also points to an important role for founder characteristics in explaining firm outcomes (De Vera et al., 2026). In my model, entrepreneurial ability is a persistent source of firm heterogeneity. It affects how quickly entrepreneurs accumulate productivity and shapes their incorporation decisions. Top income taxation then changes these firm growth paths by altering entrepreneurs' incentives to accumulate productivity and incorporate. The model therefore links heterogeneity across entrepreneurs to firm life-cycle dynamics and to the aggregate effects of tax policy.

3. Model

This section develops a structural model to study how progressive income taxation affects entrepreneurial behavior, firm dynamics, and the aggregate economy. The model builds on Quadrini (2000) and Cagetti and De Nardi (2006), but differs from the standard framework in two main ways. First, part of firm productivity is accumulated endogenously rather than evolving entirely according to an exogenous Markov process. Second, entrepreneurs can choose to incorporate their businesses, moving firms from the entrepreneurial sector to the corporate sector.

3.1. Demographics and Preferences

There is a continuum of households of measure one, indexed by $i \in [0, 1]$. Households go through two stages of life, young and old, with stochastic aging. A young household becomes old with constant probability Ω_o each period, while an old household dies with constant probability Ω_d. Following Bruggemann (2021) and Di Nola et al. (2025), households are subject to mandatory

retirement once they become old and receive a uniform retirement benefit $\overline{\tau}$ from the government, independent of their previous occupation or earnings history.

There is no bequest motive, so all bequests are accidental. When an old household dies, its assets are inherited by a newborn offspring. The newborn draws its own abilities independently of those of its parent, so there is no direct intergenerational transmission of ability. Time is discrete, with each period corresponding to one year. There is no aggregate uncertainty, although households face idiosyncratic shocks.

Following Holter et al. (2023) and Di Nola et al. (2025), I specify household preferences as follows. Workers derive utility from consumption c_t and disutility from working hours l_t, whereas entrepreneurs receive no labor disutility from running their firms (i.e., $l_t = 0$ for entrepreneurs).[3] Households' expected lifetime utility is

$$\mathbb{E}_0\Big[\sum_{t=0}^{\infty}\beta^t u(c_t, l_t)\Big] \tag{1}$$

where β is the discount factor.

3.2. Endowments and Shocks

Households are characterized by two idiosyncratic states, entrepreneurial ability z and working productivity s. A third state, business quality q, is attached to the firm rather than to the household and enters the household's state vector only when it operates a firm. To simplify notation, I suppress the household index i throughout.

Entrepreneurial Ability z. Each household draws entrepreneurial ability z from a fixed distribution $\mathcal{F}_z$ when it enters the economy. Entrepreneurial ability remains constant throughout the household's lifetime. As described in Section 3.4, a higher z allows an entrepreneur to accumulate firm productivity more quickly. I assume that entrepreneurial ability z is independent of working productivity s.

Working Productivity s. Each household draws its initial working productivity s from $\mathcal{F}_s$. Thereafter, s evolves according to a Markov process $\Pr(s' \mid s)$ for all young households, regardless of occupation. I set $\mathcal{F}_s$ equal to the stationary distribution of this process.

Business Quality q. Business quality captures idiosyncratic shocks to firm profitability, such as demand shocks, and is attached to the firm rather than to the entrepreneur. A new firm draws its initial business quality from $\mathcal{F}_q$. Once the firm is operating, q evolves according to a Markov process $\Pr(q' \mid q)$. Households that do not operate a firm do not carry a persistent q state. If a household later starts a new firm, that firm receives a fresh draw from $\mathcal{F}_q$. I set $\mathcal{F}_q$ equal to the stationary distribution of the q process. Business quality is independent of both z and s.

[3] Entrepreneurs do not supply labor to their own firms, so they use hired labor for production. This is consistent with the firm data used in the calibration, which cover employer firms only. See Section 4 for details.

3.3. Occupations and Production Technology

The model has four occupational and demographic states: young worker, young entrepreneur, young incorporation entrepreneur, and old retiree. This section describes the occupational states and the production environment.

Workers and Retirees. Young workers supply labor to a competitive spot market and earn wage income $w_t s_t l_t$, where w_t is the equilibrium wage, s_t is working productivity, and l_t is labor supply. Retirees receive the uniform retirement benefit $\overline{\tau}$ from the government.

The model has two types of firms: entrepreneurial firms and corporate firms. I interpret entrepreneurial firms as stylized counterparts of pass-through businesses and corporate firms as stylized counterparts of C corporations in the U.S. economy. I focus on three key differences between these two firm types. First, profits of entrepreneurial firms pass through to the owner and are taxed as personal income, whereas profits of corporate firms are subject to corporate income taxation. Second, entrepreneurial firms face a financing constraint tied to the entrepreneur's assets. Corporate firms have access to outside equity, and I abstract from this financing constraint for them. Third, entrepreneurial firms are closely held and managed by the entrepreneur.[4]

Entrepreneurial Firms. A household pays a sunk entry cost C_e each time it starts a new business. The cost captures expenses associated with business formation, such as registration, legal, and setup costs. Because the cost must be paid again upon reentry, it also discourages repeated entry and exit in response to unfavorable business conditions. In this model, all new businesses enter as entrepreneurial firms. Incorporation becomes available from the firm's second period onward, so this restriction only rules out incorporation at the time of entry. This is also broadly consistent with the Kauffman Firm Survey, where only around 7% of startups are organized as C corporations at founding.

Both entrepreneurial and corporate firms produce a homogeneous good using the same production technology:

$$y_t = q_t \, h_t^{1-\gamma} \left(k_t^{\alpha} \, n_t^{1-\alpha} \right)^{\gamma}, \tag{2}$$

where $\gamma \in (0,1)$ is the span-of-control parameter, capturing diminishing returns to scale at the firm level, $\alpha \in (0,1)$ is the capital share in the composite input, h_t is organizational capital, k_t is physical capital, and n_t is labor input. The production technology follows Atkeson and Kehoe (2005). The accumulation of organizational capital is described in the next section.

Incorporation Entrepreneurs. An entrepreneur who chooses to incorporate enters a one-period transition state as an incorporation entrepreneur. The incorporation decision and the valuation of the firm are described in Section 3.5.

[4]Pass-through businesses in the U.S. include sole proprietorship, partnership, and S corporation. The entrepreneurial firm in the model is a stylized representation of this sector. According to the Integrated Business Data (IBD), sole proprietorships and partnerships are the most common forms of organization, accounting for more than 80% of all businesses. Although S corporations may have multiple shareholders, they are also typically closely held: about 94% have three or fewer shareholders (U.S. Government Accountability Office, 2009). These facts motivate treating the entrepreneurial firm as an owner-managed, closely held business.

3.4. Organizational Capital Accumulation in Entrepreneurial Firms

I model the endogenous component of firm productivity as an intangible asset, denoted by h, and interpret it as organizational capital.[5] The organizational capital literature views organizational capital as a firm specific stock of knowledge embodied in the organization, including knowledge about production procedures, product development, marketing, and the organization of production itself (see, for example, Faria, 2008).[6] Organizational capital allows a firm to use its tangible inputs more effectively in production.

A central idea in this literature is that organizational capital is built through production experience. Arrow (1962) develops the learning by doing mechanism through which productive knowledge accumulates with experience, while Rosen (1972), Atkeson and Kehoe (2005), and Lev and Radhakrishnan (2005) view organizational capital as firm specific capital that is jointly produced with output. Following this idea, I assume that entrepreneurs start their businesses with the same initial stock of organizational capital, h_0, and accumulate it as they produce. I specify the law of motion for organizational capital as

$$h_{t+1} = (1 - \delta_h)\, h_t + z\, y_t^{\psi}, \tag{3}$$

where δ_h is the depreciation rate of organizational capital and $\psi \in (0, 1)$.

The key feature of (3) is that organizational capital accumulation depends on both current production and entrepreneurial ability. A higher z makes the entrepreneur more effective at turning production experience into organizational capital. Thus, for a given level of output, firms run by higher z entrepreneurs accumulate organizational capital more quickly. This is consistent with recent empirical evidence suggesting that the characteristics of entrepreneurs are important for firm performance (De Vera et al., 2026).

3.5. Incorporation and the Corporate Sector

An entrepreneur can choose to continue operating her entrepreneurial firm, shut down the business and become a worker, or incorporate the firm. Incorporation involves paying a fixed incorporation cost C and selling the firm to a competitive mutual fund, which then owns the firm. The incorporation cost captures legal, accounting, and other setup costs associated with moving the firm into the corporate sector. In return, the entrepreneur receives the firm value, denoted by Π, and pays capital gains tax at rate τ_k on this value. The firm value is the present value of the future after tax profits generated by the corporate firm. Because the mutual fund operates competitively, the purchase price is equal to this market value.

In this model, incorporation means that the entrepreneur no longer owns the firm, and the firm moves into the corporate sector. The founder then exits management and leaves the firm. In

[5] Other forms of entrepreneurial intangible assets have also been studied in the literature. For example, Bhandari and McGrattan (2021) model sweat equity accumulated through entrepreneurial time and effort. In this paper, I instead interpret h as organizational capital.

[6] Related work on organizational capital includes Arrow (1962), Rosen (1972), Prescott and Visscher (1980), Tomer (1987), Ericson and Pakes (1995), and Eisfeldt and Papanikolaou (2013), among others.

reality, founders can sell only part of their ownership and retain a stake in the firm. I instead model entrepreneurial and corporate firms as two distinct ownership regimes. This is a stylized way to represent the ownership transition.[7]

As discussed in Section 3.4, organizational capital is firm specific and embedded in the organization. I therefore assume that the organizational capital accumulated before incorporation remains with the firm and does not go with the entrepreneur when she incorporates. Incorporation thus does not create a new firm or reset its organizational capital. The same is true for business quality q. After incorporation, q continues to evolve according to the same Markov process $\Pr(q' \mid q)$ rather than being redrawn from the initial distribution.

Incorporation changes the ownership of the firm but not its underlying production technology. I therefore assume that corporate firms use the same production function (2) as entrepreneurial firms. Organizational capital accumulation, however, may change after the ownership transition. I capture this by allowing the efficiency of organizational capital accumulation to change after incorporation. In particular, organizational capital in a corporate firm evolves according to

$$h' = (1 - \delta_h)\, h + \theta\, z\, y^{\psi}, \qquad \theta > 0 \tag{4}$$

where θ captures how organizational capital accumulation changes after incorporation. If $\theta = 1$, organizational capital accumulates in the same way as before incorporation. If θ is above or below one, the accumulation path changes accordingly. The value of θ is determined in the calibration.

Given these post incorporation dynamics, the value of the corporate firm can be written as

$$\Pi(h, q, z) = \max_{k,n} \left\{ (1 - \tau_c)\, \pi^c + \frac{1 - D}{1 + r}\, \mathbb{E}\big[\Pi(h', q', z) \,\big|\, q\big] \right\}, \tag{5}$$

where

$$\begin{aligned} \pi^c &= q h^{1-\gamma} (k^{\alpha} n^{1-\alpha})^{\gamma} - wn - (r + \delta)k, \\ h' &= (1 - \delta_h)\, h + \theta\, z\, y^{\psi}. \end{aligned}$$

Corporate profits are taxed at the flat corporate income tax rate τ_c. Together with the capital gains tax paid by the founder on Π, corporate income in this model is subject to double taxation, as in the U.S.[8] In each period, each incorporated firm faces an exogenous death shock with probability D, which gives corporate firms a finite expected lifetime.

[7] Some empirical evidence is consistent with this modeling choice. For example, Wasserman (2003) finds that the attainment of firm milestones, such as raising capital from outside investors, tends to trigger founder-manager succession. Additional evidence comes from IPOs, which are a narrower event than incorporation in this model. Broughman and Fried (2020) document that among firms that complete an IPO, 60% of founders are no longer CEO at the time, and of those who remain, only half survive as CEO for another three years.

[8] In the U.S., income earned by a C corporation is subject to two layers of taxation. The corporation first pays corporate income tax on its profits. The owners then face a second layer of taxation, in one of two ways: as a dividend tax when the profits are paid out, or as a capital gains tax when they sell their shares, whose price reflects the value of those profits. The model uses the second route. Since the firm value Π is the present value of the firm's future profit flow net of corporate income tax, taxing Π provides a second layer of taxation on the value generated by those profits.

3.6. The Mutual Fund and Firm Financing

There is a competitive mutual fund that channels household savings to firms. Households cannot hold corporate equity directly. Each period, households deposit their savings with the fund and earn a risk-free return r. The fund uses household savings to supply physical capital to both entrepreneurial and corporate firms and also holds the equity of all corporate firms. When an entrepreneur incorporates, she sells the firm to the mutual fund at the market value Π, as described in Section 3.5. The founder therefore realizes the value of her ownership at incorporation, while the mutual fund holds the corporate equity thereafter.

The fund rents physical capital to both entrepreneurial and corporate firms at the rental rate r. The fund also earns the return on its corporate equity portfolio. Because it holds a continuum of corporate firms, idiosyncratic business quality shocks (q) and corporate death shocks wash out, so the return on this portfolio is deterministic. No arbitrage then requires the corporate portfolio to earn the same return r as physical capital. The fund can therefore pay households the risk-free return r on their deposits. Moreover, competition implies that the fund earns zero economic profit. When the fund acquires a corporate firm, it pays the founder the market value Π, which reflects the value of the firm's future after-tax profits. The returns subsequently generated by the fund's assets are passed on to household savers, so the fund retains no additional profit from intermediation.

Households cannot borrow to finance consumption, so savings are non-negative each period. Firms rent physical capital from the mutual fund within the period and return it after production. Entrepreneurial firms are subject to a collateral constraint on capital:

$$k_t \leq \eta \, a_t, \tag{6}$$

where a_t denotes the entrepreneur's personal assets. Corporate firms, on the other hand, are not subject to this financial constraint.

3.7. Government

The government collects personal income, corporate income, and capital gains taxes. Personal income is subject to a progressive tax schedule, denoted by $\mathbb{T}(y^{inc})$. Corporate profits are taxed at the flat rate τ_c, and capital gains realized upon incorporation are taxed at the flat rate τ_k. Tax revenues finance wasteful government consumption G and a uniform retirement benefit $\overline{\tau}$ paid to all retirees. The government budget is balanced in equilibrium.

3.8. Household Problems

This section formalizes the optimization problems of the four household types. The state of a young worker is (a, s, z), while the state of a young entrepreneur is (a, s, h, q, z). A young incorporation entrepreneur has the same state variables because the value of the firm sold at incorporation depends on (h, q, z). The state of an old retiree is given by assets a.

The timing within a period is as follows. A household enters period t with the occupation chosen at the end of the previous period. It then observes its current working productivity s_t

and, if it operates a firm, its current business quality q_t. The household then makes its relevant within period decisions. For an entrepreneur, production takes place and organizational capital evolves to h_{t+1}. At the end of the period, conditional on remaining young, the household chooses its occupation for period $t+1$. This choice is made before s_{t+1} and q_{t+1} are realized. Working productivity is then drawn from $\Pr(s_{t+1} \mid s_t)$. For a continuing firm, business quality is drawn from $\Pr(q_{t+1} \mid q_t)$, while a newly entered firm receives a fresh draw from $\mathcal{F}_q$.

3.8.1. Old Retirees

All old households are retired, and their state is fully described by assets a. The only uncertainty they face is the constant probability of death Ω_d. Households have no bequest motive, so any assets left at death are accidental bequests. An old retiree receives the uniform retirement benefit $\overline{\tau}$ and capital income ra. Both are subject to the personal income tax schedule $\mathbb{T}(\cdot)$.

$$
\begin{aligned}
& V^r(a) = \max_{c,a'} \; u(c,0) + \beta(1-\Omega_d)V^r(a') \qquad (7)\\
& s.t.\\
& c + a' = y^{inc} + a - \mathbb{T}(y^{inc})\\
& y^{inc} = \overline{\tau} + ra\\
& a' \geq 0
\end{aligned}
$$

3.8.2. Young Worker

Young workers choose labor supply l, consumption c, and savings a' to maximize their lifetime utility. With probability Ω_o, the worker becomes old and retires next period. Conditional on remaining young, the worker chooses at the end of the period whether to continue as a worker or pay the sunk entry cost C_e to become an entrepreneur. This choice is made before next period's idiosyncratic shocks are realized. The worker's problem is

$$
\begin{aligned}
V^w(a,s,z) = \max_{l,c,a'} \; & u(c,l) + \beta(1-\Omega_o)\max\left\{\mathbb{E}[V^w(a',s',z)|s],\; \mathbb{E}[V^e(a'-C_e,s',h_0,q',z)|s]\right\}\\
& + \beta\Omega_o V^r(a') \qquad (8)\\
s.t.\\
c + a' &= y^{inc} + a - \mathbb{T}(y^{inc})\\
y^{inc} &= wsl + ra\\
a' &\geq 0
\end{aligned}
$$

In the entrepreneurship option, s' is drawn from $\Pr(s' \mid s)$ and the new firm receives a fresh draw q' from $\mathcal{F}_q$. Entry requires $a' \geq C_e$. If the worker enters entrepreneurship, the entry cost C_e is paid at the end of the current period, so the entrepreneur begins the following period with assets $a' - C_e$ and organizational capital h_0.

3.8.3. Young Entrepreneur

Entrepreneurs choose capital input k, hired labor n, consumption c, and savings a'. Conditional on remaining young, at the end of the period they choose whether to continue operating the firm,

shut it down and become a worker, or incorporate it. The entrepreneur's problem is

$$V^e(a,s,h,q,z) = \max_{k,\,n,\,c,\,a'} u(c,0) + \beta\Omega_o\, V^r(a') + \beta(1-\Omega_o)\max\{\mathbb{E}\big[V^w(a',s',z)\,\big|\,s\big]\,, \\ \mathbb{E}\big[V^e(a',s',h',q',z)\,\big|\,s,q\big]\,,\ \mathbb{E}\big[V^i(a',s',h',q',z)\,\big|\,s,q\big]\} \tag{9}$$

$$\begin{aligned} &s.t. \\ &c + a' = a + y^{inc} - \mathbb{T}(y^{inc}) \\ &y^{inc} = qh^{1-\gamma}(k^\alpha n^{1-\alpha})^\gamma - wn - \delta k - r(k-a) \\ &h' = (1-\delta_h)\,h + z\left[qh^{1-\gamma}(k^\alpha n^{1-\alpha})^\gamma\right]^\psi \\ &k \le \eta a, \quad a' \ge 0 \end{aligned}$$

The entrepreneur's income includes output net of the wage bill, capital depreciation, and net capital rental payments. The entire income y^{inc} is subject to the progressive personal income tax $\mathbb{T}(\cdot)$, consistent with the pass-through tax treatment.

3.8.4. Incorporation Entrepreneur

When an entrepreneur chooses to incorporate, she enters a one-period incorporation state. During this period, the entrepreneur receives the firm value $\Pi(h,q,z)$, pays the fixed incorporation cost C and capital gains tax at rate τ_k, and earns capital income ra on personal assets. Her occupation choice for the next period is between becoming a worker and starting a new business. Because organizational capital h is embedded in the sold firm and is not carried by the founder, if the founder later re-enters entrepreneurship, the new firm therefore starts with organizational capital h_0. The incorporation entrepreneur's problem is

$$\begin{aligned} &V^i(a,s,h,q,z) = \max_{c,a'}\ u(c,0) + \beta\Omega_o V^r(a') \\ &+ \beta(1-\Omega_o)\max\Big\{\mathbb{E}[V^w(a',s',z)|s],\ \mathbb{E}[V^e(a'-C_e,s',h_0,q',z)|s]\Big\} \\ &s.t. \\ &c + a' = y^{inc} + a + (1-\tau_k)\Pi(h,q,z) - C - \mathbb{T}(y^{inc}) \\ &y^{inc} = ra, \quad a' \ge 0 \end{aligned} \tag{10}$$

The entrepreneur's taxable income in the incorporation stage consists only of capital income, $y^{inc} = ra$. The proceeds from the sale of the firm, $(1-\tau_k)\Pi(h,q,z)$, are received net of the capital gains tax and enter the budget constraint separately from ordinary income subject to $\mathbb{T}(\cdot)$.

The founder and the firm follow separate paths during the incorporation period. Suppose an entrepreneur operates an entrepreneurial firm in period t. At the end of this period, she makes her occupational choice for the next period and decides to incorporate. By that point, production has taken place and organizational capital has evolved to h_{t+1}, while q_{t+1} has not yet been realized. In period $t+1$, the founder enters the one-period incorporation state described above. At the same time, the firm is already operating in the corporate sector, with organizational capital h_{t+1} and business quality $q_{t+1} \sim \Pr(q_{t+1} \mid q_t)$. Thus, incorporation separates the subsequent path of

the founder from that of the firm: the firm continues with its accumulated organizational capital and business quality, while the founder no longer operates it. Incorporation therefore changes the ownership of the existing firm rather than creating a new firm.

At the end of the incorporation period, the founder chooses between wage work and starting a new business in the next period. This creates a channel for serial entrepreneurship in the model. Although organizational capital is firm specific and a new business must start again from h_0, entrepreneurial ability z is a permanent characteristic of the household. A high z household can therefore build organizational capital relatively quickly in a new business, making re-entry into entrepreneurship attractive.

3.9. Equilibrium

The state space of the model economy comprises asset holdings $a \in \mathcal{A}$, organizational capital $h \in \mathcal{H}$, entrepreneurial ability $z \in \mathcal{Z}$, working productivity $s \in \mathcal{S}$, business quality $q \in \mathcal{Q}$, and occupation $o \in \mathcal{O}$. There are four occupation states, $\mathcal{O} = \{YW, YE, YI, OR\}$, corresponding to young workers, young entrepreneurs, young incorporation entrepreneurs, and old retirees, respectively. The complete household state space is $\Xi = \mathcal{A} \times \mathcal{Z} \times \mathcal{S} \times \mathcal{Q} \times \mathcal{H} \times \mathcal{O}$, with state vector $\xi \in \Xi$. The corporate sector state space is $\Xi^C = \mathcal{Q} \times \mathcal{H} \times \mathcal{Z}$, with state vector $\xi^c \in \Xi^C$. A stationary equilibrium of the model is defined by

1. Prices: interest rate r and wage w;
2. Value functions: $V^r(a)$, $V^w(a, s, z)$, $V^e(a, s, h, q, z)$, $V^i(a, s, h, q, z)$, and corporate firm value $\Pi(h, q, z)$;
3. Household policy functions: consumption $c(\xi)$, labor supply $l(\xi)$, savings $a'(\xi)$, and occupational choices;
4. Capital input $k(\xi)$, labor input $n(\xi)$, and implied organizational capital $h'(\xi)$ for entrepreneurs;
5. Corporate firm policy functions: capital $k(\xi^c)$, labor $n(\xi^c)$, and implied organizational capital $h'(\xi^c)$
6. An invariant distribution of households Λ over Ξ;
7. An invariant distribution of corporate firms Λ^C over Ξ^C;

such that the following conditions hold:

1. Household optimality. Given prices and taxes, the households' decision rules and value functions solve their respective dynamic programming problems.
2. Corporate firm optimality. Given prices (r, w) and corporate tax rate τ_c, each corporate firm chooses capital and labor to maximize firm value $\Pi(h, q, z)$.
3. Government budget balance. The government balances its budget period by period. Tax revenues finance wasteful government consumption G and uniform retirement benefits $\overline{\tau}$ paid to all retirees:

$$\int \mathbb{T}(y^{inc})\, d\Lambda + \int \tau_k\, \Pi(h, q, z) \cdot \mathbf{1}_{\{o=YI\}}\, d\Lambda + \tau_c \int \pi^c\, d\Lambda^C = G + \overline{\tau} \cdot \mu_{OR}, \tag{11}$$

where π^c is the corporate operating profit and μ_{OR} is the measure of old retirees in the stationary distribution.

4. Goods market clearing. Let $\chi_e(\xi)$ equal one if a household starts a new business in the following period and zero otherwise. Aggregate output from entrepreneurial and corporate firms, net of fixed entry and incorporation costs, equals aggregate household consumption, physical capital depreciation, and wasteful government consumption.

$$\int y \, d\Lambda + \int y^c \, d\Lambda^C - \int C \cdot \mathbf{1}_{\{o=YI\}} \, d\Lambda - \int C_e \cdot \chi_e(\xi) \, d\Lambda = \int c \, d\Lambda + \delta(K^e + K^c) + G \tag{12}$$

5. Labor market clearing. The aggregate supply of efficiency units of labor from young workers to the spot market must equal the aggregate demand for hired labor from entrepreneurial firms and incorporated firms.
6. Capital market clearing. The mutual fund's total assets consist of three components: capital rented to entrepreneurial firms, physical capital deployed in the corporate sector, and the equity value of incorporated firms. Household savings finance all three components.
7. Mutual fund zero profit.
8. Time-invariant distribution.

$$(\Lambda, \Lambda^C) = \Gamma(\Lambda, \Lambda^C), \tag{13}$$

where Γ is the law of motion implied by the policy functions, the exogenous shocks, aging and death, and the replacement of each deceased household by a newborn who inherits its assets.

4. Calibration and Model Performance

This section describes how I calibrate the model to the U.S. economy using data from 2013 to 2016, a period with a relatively stable federal tax environment. I follow a two-step calibration procedure. I calibrate a subset of parameters externally using estimates from the literature or the U.S. tax code. These parameters are reported in Tab 1. The remaining 15 parameters are jointly calibrated to match key features of the U.S. economy and are reported in Tab 2. I then evaluate the model's performance using both targeted and untargeted moments.

I use three main data sources. The Survey of Consumer Finances (SCF) provides information on household income, wealth, and entrepreneurship. The Business Dynamics Statistics (BDS), constructed from the Longitudinal Business Database (LBD), provides information on firm dynamics, including firm exit and employment by firm age. Since the BDS does not provide information on firms' legal form of organization, I use the Statistics of U.S. Businesses (SUSB) to construct moments related to C corporations. All firms in my model are employer firms, consistent with the firm population covered by the BDS and SUSB. Accordingly, I restrict entrepreneurs in the SCF to business owners who actively manage businesses with hired employees.

Table 1. Externally Calibrated Parameters

Parameter	Description	Value	Source
Demographics and Preferences			
Ω_o	Probability of retiring (45-year working life)	$1/45$	Assumption
Ω_d	Probability of dying (15-year retirement)	$1/15$	Assumption
σ_1	Relative risk aversion	1.5	Bruggemann (2021)
σ_2	Inverse Frisch elasticity	1.7	Bruggemann (2021)
Working Productivity			
ρ_s	Persistence of working productivity	0.90	Guvenen et al. (2023)
σ_s	Std. dev. of working productivity innovations	0.20	Guvenen et al. (2023)
Production Technology			
h_0	Initial organizational capital	1	Normalization
γ	Span-of-control parameter	0.85	Atkeson and Kehoe (2005)
α	Capital share	0.36	Cooley and Prescott (1995)
δ	Physical capital depreciation rate	0.05	Guvenen et al. (2023)
δ_h	Organizational capital depreciation rate	0.15	Eisfeldt and Papanikolaou (2013)
Financial Frictions			
η	Collateral constraint parameter	1.5	Kitao (2008)
Corporate Sector			
D	Exogenous corporate exit probability	0.09	Dyrda and Pugsley (2022)
Government			
$\overline{\tau}$	Retirement benefit (40% of avg. labor income)	$0.4\,\bar{y}$	Bruggemann (2021)
λ	Income tax progressivity	0.14	Gao and Zhang (2024)
τ_k	Capital gains tax rate	0.18	U.S. Treasury
τ_c	Corporate income tax rate	0.20	U.S. Treasury

Table 2. Internally Calibrated Parameters

Parameter	Description	Value
Demographics and Preferences		
β	Discount factor	0.96
χ	Weight on labor disutility	11.72
Entrepreneurial Ability		
$z_{\min}$	Lower bound of entrepreneurial ability	0.93
ζ	Pareto shape parameter	3.36
Working Productivity		
s_H	Top working productivity state	15.13
p_H	Probability of entering the top productivity state	0.04
p_M	Probability of exiting the top productivity state	0.26
Business Quality		
μ_q	Level of business quality	0.06
ρ_q	Persistence of business quality	0.71
σ_q	Std. dev. of business quality innovations	0.23
Organizational Capital Accumulation		
ψ	Curvature of organizational capital accumulation	0.42
θ	Corporate organizational capital accumulation efficiency	0.27
Fixed Costs		
C_e	Entrepreneurship entry cost	1.02
C	Incorporation cost	9.90
Taxation		
τ_y	Income tax level	0.99

4.1. Calibration Strategy

I calibrate the internal parameters by minimizing the distance between model moments and their counterparts in the data. The targets include both household-level moments, such as income and wealth distributions, and firm-level moments that capture firm dynamics. Although all parameters jointly affect the targeted moments, I highlight below which moments are most informative about each parameter.

4.1.1. Demographics and Preferences

A young household becomes old with constant probability Ω_o, and an old household dies with constant probability Ω_d each period. One period corresponds to one year. I assume an expected working life of 45 years, corresponding to ages 20 to 65, and an expected retirement period of 15 years, corresponding to ages 65 to 80. Accordingly, I set $\Omega_o = 1/45$ and $\Omega_d = 1/15$.

The utility function is of CRRA type in consumption and additively separable in consumption and labor:

$$u(c,l) = \frac{c^{1-\sigma_1}}{1-\sigma_1} - \chi \frac{l^{1+\sigma_2}}{1+\sigma_2}, \tag{14}$$

The coefficient of relative risk aversion σ_1 is set to be 1.5, a standard value in the macroeconomic literature (for example, Bruggemann, 2021). The parameter σ_2 governs the Frisch elasticity of labor supply, which is $1/\sigma_2$. Following Bruggemann (2021), I set $\sigma_2 = 1.7$, corresponding to a Frisch elasticity of approximately 0.59. The weight on labor disutility, χ, is calibrated so that the average hours worked by workers in the model is 0.38 (Dyrda and Pugsley, 2022). Finally, I calibrate the discount factor β to match a physical capital-to-output ratio of 3.33, as reported by Di Nola et al. (2025).

4.1.2. Endowments and Shocks

There are three types of endowments and shocks in the model: entrepreneurial ability z, working productivity s, and business quality q. Households draw their entrepreneurial ability z when they first enter the economy, and it remains fixed throughout their lives. Working productivity s and firm specific business quality q evolve stochastically over time.

Entrepreneurial Ability z. It follows a type-I Pareto distribution with scale parameter $z_{\min}$ and shape parameter ζ. Its density and survival functions are given by

$$f_Z(z) = \frac{\zeta\, z_{\min}^{\zeta}}{z^{1+\zeta}}, \qquad S_Z(z) \equiv \Pr(Z > z) = \left(\frac{z_{\min}}{z}\right)^{\zeta}, \qquad z \geq z_{\min}. \tag{15}$$

The Pareto density is monotonically decreasing, so most households draw relatively low entrepreneurial ability and tend to remain workers, while a small fraction draw very high ability.

The shape parameter ζ governs the thickness of the right tail of the entrepreneurial ability distribution. A smaller ζ implies a thicker right tail, which affects wealth concentration at the top. I therefore calibrate ζ to match the top 1% wealth share. The scale parameter $z_{\min}$ determines the lower bound of entrepreneurial ability. Since z enters the accumulation equation for organizational capital h, it governs how quickly entrepreneurs can build organizational capital and increase future profitability. A lower $z_{\min}$ makes entrepreneurship less attractive for most households, leading more of them to remain workers. I calibrate $z_{\min}$ to match the fraction of entrepreneurs in working-age population, which is 0.08 in the SCF.

Working Productivity s. I assume that working productivity follows an AR(1) process given by

$$\log s_t = \rho_s \log s_{t-1} + \varepsilon_t^s, \qquad \varepsilon_t^s \sim N(0, \sigma_s^2). \tag{16}$$

The annual persistence of working productivity, ρ_s, and the standard deviation of productivity shocks, σ_s, are set to 0.9 and 0.2, respectively, following Guvenen et al. (2023). I discretize this continuous AR(1) process into five grid points, $s \in \{s_1, \ldots, s_5\}$. Since this paper studies top income taxation, it is important for the model to capture the income concentration at the top, as well as the composition of workers and entrepreneurs in the top income group. Following Castaneda et al. (2003), I introduce an additional high labor productivity state, denoted by s_H.

Households in productivity state s_5 can transition to s_H with probability p_H. Once in state s_H, they face a probability p_M of returning to the middle productivity state s_3. This extension introduces three additional parameters, (s_H, p_H, p_M), which govern the level of high labor productivity and the transitions into and out of this state. I jointly calibrate these parameters to match three top income moments: (i) the top 1% income share, (ii) the share of entrepreneurs among the top 1% income group, and (iii) the share of top 1% income held by entrepreneurs. All three moments are computed using the SCF.

Business Quality q**.** I assume that business quality follows an AR(1) process given by

$$\log q_t = \mu_q + \rho_q \log q_{t-1} + \varepsilon_t^q, \qquad \varepsilon_t^q \sim N(0, \sigma_q^2). \tag{17}$$

The parameter μ_q determines the level of business quality, which directly affects firm production and profitability. I therefore calibrate μ_q to match the share of income earned by entrepreneurs in the SCF. The other two parameters, ρ_q and σ_q, govern the persistence of business quality and the dispersion of business quality shocks, respectively. They are jointly calibrated with ψ, which governs organizational capital accumulation, to match firm dynamics moments. I discuss their calibration in the next subsection.

4.1.3. Production Technology

There are two production sectors in this economy: the entrepreneurial sector and the corporate sector. Entrepreneurial firms pay a fixed entry cost C_e and start with the same initial level of organizational capital h_0. Entrepreneurs produce, accumulate organizational capital, and make incorporation decisions. Both physical and organizational capital depreciate over time.

I calibrate the fixed entry cost C_e to match the average age at first entry into entrepreneurship, which ranges from 32 to 36 in Hincapie (2020). I target an average entry age of 33 in the model. All new firms start with the same initial level of organizational capital, h_0, which I normalize to 1.

I adopt the same production function with organizational capital as in Atkeson and Kehoe (2005):

$$y = q\, h^{1-\gamma} \left(k^{\alpha}\, n^{1-\alpha}\right)^{\gamma}$$

Following Atkeson and Kehoe (2005), I set the span-of-control parameter $\gamma = 0.85$, implying an organizational capital share of $1 - \gamma = 0.15$. This is consistent with estimates in the literature. For example, Corrado et al. (2009) estimate that income accruing to intangible capital accounts for approximately 15% of total income in the U.S. non-farm business sector. I set $\alpha = 0.36$, a standard value in the literature (Cooley and Prescott, 1995). Together, these parameter values imply an organizational capital share of 0.15 and a physical capital share of $0.85 \times 0.36 \approx 0.31$. These values are also consistent with Bhandari and McGrattan (2021), who estimate an intangible capital (sweat capital) share of 0.15 and a fixed asset share of 0.30 for U.S. private businesses. Finally, I set the physical capital depreciation rate $\delta = 0.05$, following Guvenen et al. (2023).

Organizational capital accumulates according to

$$h' = (1 - \delta_h)\, h \; + z\, y^{\psi}$$

I set the depreciation rate of organizational capital to $\delta_h = 0.15$, following Eisfeldt and Papanikolaou (2013). The parameter ψ, together with the two parameters governing the business quality process (ρ_q and σ_q), plays an important role in determining firm dynamics.

The key intuition is that business quality shocks, together with organizational capital accumulation, can generate the "up-or-out" dynamics of young firms documented by Fort et al. (2013), Haltiwanger (2012), and Haltiwanger et al. (2013). Young firms have not yet accumulated much organizational capital, making them more vulnerable to negative business quality shocks and more likely to exit. As surviving firms age, they accumulate organizational capital, which helps them withstand negative shocks and lowers their exit rates. These dynamics depend on how quickly firms accumulate organizational capital and how business quality evolves over time.

I jointly calibrate ρ_q, σ_q, and ψ to three sets of BDS moments: the overall exit rate, the survival rates of young (ages 0–2) and mature (ages 11 and above) firms, and the employment shares of young and mature firms. The dispersion parameter σ_q affects the likelihood of large negative business quality shocks. Since these shocks can trigger exit among firms of all ages, the overall exit rate provides important information for calibrating σ_q. The persistence parameter ρ_q governs how long a firm remains in a bad quality state. Since organizational capital helps firms withstand negative shocks and tends to increase as surviving firms age, the persistence of business quality affects the exit rates of young and mature firms differently. I therefore use their survival rates to help calibrate ρ_q.

Finally, ψ governs how quickly organizational capital builds up, and hence how fast surviving firms grow. Given that the survival rates help discipline the number of young and mature firms, their employment shares provide information about their relative sizes and help calibrate ψ. Since the BDS covers employer firms regardless of their legal form of organization, I compute the corresponding firm dynamics moments in the model using firms from both the entrepreneurial and corporate sectors.

4.1.4. Incorporation

Entrepreneurs can choose to incorporate their firms by paying a fixed cost C. Corporate firms continue to produce using the same production function as entrepreneurial firms and also accumulate organizational capital. They face an exogenous death probability D each period. I calibrate the incorporation cost C to match the share of C corporations among all employer firms, which is 18% in the SUSB. I set the exogenous death probability to $D = 0.09$, following Dyrda and Pugsley (2022), who report an annual exit rate of 8–10% for C corporations around 2012.

Corporate sector firms accumulate h as

$$h' = (1 - \delta_h)\, h + \theta\, z\, y^{\psi}, \qquad \theta > 0$$

I calibrate θ to match the share of employment by C corporations, which is 52% in the SUSB. The incorporation cost C mainly affects how many firms choose to incorporate. Given the number of corporate firms, their employment share provides information about their size, which is influenced by θ. A higher θ allows corporate firms to accumulate organizational capital more efficiently and grow larger.

In the calibration, I find that θ is less than one. This implies that, given z and y, the efficiency of organizational capital accumulation is lower after incorporation. One possible explanation is that when a founder exits management, her knowledge about the firm may not be fully transferred to the new management. Part of what kept the firm improving was her own judgment, contacts, and firm-specific knowledge, which may be difficult to transfer to someone else. A similar effect may arise even if the founder remains with the firm after incorporation. In practice, founders may remain with their firms after bringing in outside shareholders. However, they now share control with other shareholders and a board of directors, which may limit their ability to make decisions about how to operate and grow the firm as they did before. This may also affect the efficiency of organizational capital accumulation. My model does not explicitly capture either of these channels. Instead, θ is a reduced-form parameter that captures the change in organizational capital accumulation efficiency after incorporation.

4.1.5. Financing and Government

Entrepreneurs face the collateral constraint $k \leq \eta a$. I set $\eta = 1.5$, following Kitao (2008) and Bruggemann (2021).

The expenditure side of the government budget consists of wasteful government spending and total retirement benefits paid out to retirees. On the revenue side, there are three taxes: a progressive income tax, a flat capital gains tax, and a flat corporate income tax. Government spending G is determined to balance the government budget in the benchmark economy. Following Bruggemann (2021), I set the retirement benefit $\overline{\tau}$ to 40% of average labor income.

I adopt the income tax function of Benabou (2002) and adjust it to study the top tax rate, as in Di Nola et al. (2025) and Ge (2026):

$$\mathbb{T}(y) = \begin{cases} y - \tau_y \, y^{1-\lambda} & y \leq y_H \\ y_H - \tau_y \, y_H^{1-\lambda} + \tau_H (y - y_H) & y > y_H \end{cases} \tag{18}$$

The parameter τ_y controls the overall level of income taxation, while λ governs the progressivity of the tax schedule. The income tax function is nonlinear below the income threshold y_H and linear above it. The threshold y_H is the income cutoff for the top 1% taxable income, and τ_H is the top marginal income tax rate.

I calibrate τ_y to match the federal individual income tax revenue-to-GDP ratio, which averaged approximately 8% during 2013–2016, according to the Office of Management and Budget (OMB). I set the progressivity parameter $\lambda = 0.14$, following Gao and Zhang (2024). The threshold y_H is determined endogenously in the benchmark equilibrium. To ensure continuity of the marginal income tax rate at y_H in the benchmark economy (see, e.g., Di Nola et al., 2025 and Ge, 2026), the top marginal tax rate is calculated using the following equation:

$$\tau_H = 1 - \tau_y (1 - \lambda) y_H^{-\lambda}. \tag{19}$$

This gives a benchmark top marginal tax rate of $\tau_H = 0.37$, close to the U.S. federal top marginal income tax rate of 0.39 during this period. In the policy experiments, I vary τ_H while holding y_H fixed at its benchmark value.

I set the corporate income tax rate to $\tau_c = 0.20$, based on the average effective federal corporate tax rate reported by the U.S. Department of the Treasury. The capital gains tax rate is set to $\tau_k = 0.18$, based on the average capital gains tax rate estimated by the U.S. Treasury for 2014.

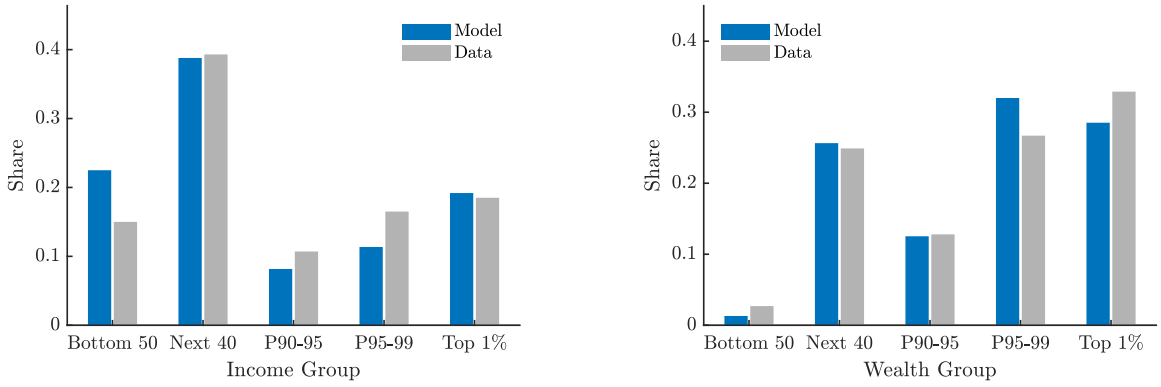
4.2. Model Performance

I evaluate the performance of the benchmark economy by comparing model generated moments with their counterparts in the data, including moments not directly targeted in the calibration. Table 3 compares the targeted moments in the data with their model counterparts. Overall, the model provides a good fit to these moments, including aggregate statistics, income and wealth concentration, entrepreneurship, and firm dynamics.

Table 3. Targeted Moments

Moment	Model	Target
Aggregates		
K/Y ratio	3.15	3.33
Average hours worked	0.37	0.38
Income tax revenue-to-GDP ratio	0.07	0.08
Entrepreneurs		
Share of entrepreneurs in the working age population	0.09	0.08
Share of income earned by entrepreneurs	0.20	0.21
Share of entrepreneurs among the top 1% income group	0.47	0.49
Share of top 1% income held by entrepreneurs	0.55	0.56
Avg. age at first entry into entrepreneurship	32	33
Inequality		
Top 1% wealth share	0.29	0.33
Top 1% income share	0.19	0.19
Firm Dynamics		
Overall firm exit rate	0.07	0.07
Survival rate of young firms (ages 0–2)	0.82	0.82
Survival rate of mature firms (ages $11+$)	0.94	0.95
Employment share of young firms	0.07	0.06
Employment share of mature firms	0.79	0.81
Corporate Sector		
Share of C corporations among employer firms	0.15	0.18
Employment share of C corporations	0.55	0.52

Figure 1 compares the income and wealth distributions in the model and the data. Although I only target the top 1% income and wealth shares in the calibration, the model also captures the overall patterns of both distributions. In particular, wealth is more concentrated than income in both the model and the data.



(1) Income Distribution (2) Wealth Distribution

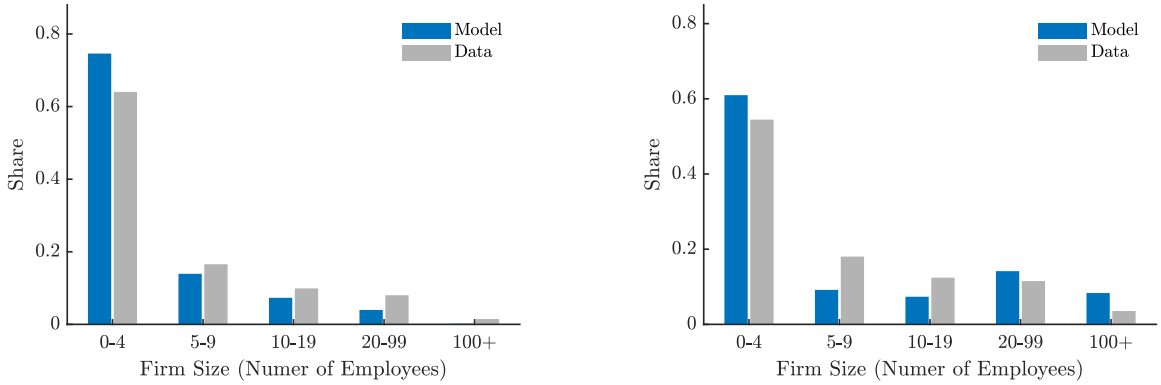
Figure 1. Inequality

Notes: The data moments are computed from the SCF. Only the top 1% income and wealth shares are targeted in the calibration. The shares for all other groups are untargeted moments.

Table 4 provides an additional check on the income differences between entrepreneurs and workers. The mean income ratio of entrepreneurs to workers is 2.73 in the data, compared with 2.33 in the model. The model also generates a median income ratio of 1.43, compared with 1.79 in the data. The median ratio is particularly informative. Although I target the share of total income earned by entrepreneurs, this share could be driven by a small number of very high income entrepreneurs. The median ratio provides an additional check on whether the model also captures the income difference between typical entrepreneurs and workers.

Table 4. Untargeted Moments

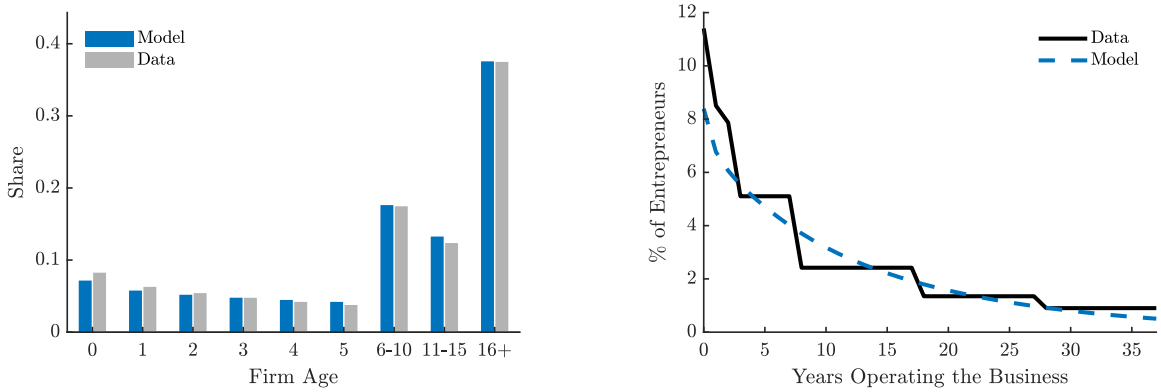
Moment	Model	Data	Source
Share of output by pass-through firms	0.43	0.40	Di Nola et al. (2025)
Avg. business income ratio: pass-through to c-corps	0.15	0.20	Dyrda and Pugsley (2022)
Mean income ratio: entrepreneurs to workers	2.33	2.73	SCF
Median income ratio: entrepreneurs to workers	1.43	1.79	SCF



(1) Pass-through Firms (2) Corporate Firms

Figure 2. Firm Size Distribution

Notes: The data moments are computed from the SUSB. Firm size is measured by the number of employees. The figure shows the distribution of firms across size categories, separately for pass-through firms and C corporations.



(1) Firm Share by Age (2) Entrepreneurs' Business Tenure

Figure 3. Firm Dynamics

Notes: The left panel shows the distribution of firms by age using BDS data. Since the BDS does not distinguish firms by legal form of organization, I compute the corresponding model moments using firms from both the entrepreneurial and corporate sectors. The right panel shows the distribution of entrepreneurs by the number of years they have operated their current business. The data are from the 2007 Survey of Business Owners (SBO) microsample.

For the corporate sector, I only target the share of C corporations and their employment share in the calibration. Figure 2 compares the firm size distributions of pass-through firms and C corporations in the data with those of entrepreneurial and corporate firms in the model, respectively. The model captures the overall patterns of both distributions. In particular, the firm size distribution of C corporations depends on which firms choose to incorporate and how they grow afterward. Matching this distribution provides an additional check on these two mechanisms.

I further examine two untargeted moments reported in Table 4. The output share of pass-through firms is 0.40 in the data and 0.43 in the model. Matching the employment shares of the two sectors does not guarantee that the model also captures their relative output, which depends on capital intensity and productivity. The ratio of average business income of pass-through firms to that of C corporations is 0.20 in the data and 0.15 in the model. This provides an additional check on the relative business income of firms in the two sectors.

The left panel of Figure 3 compares the firm age distributions in the model and the data. Although I only target the survival rates of young and mature firms in the calibration, the model also captures the overall firm age distribution. In the model, firm survival depends on business quality shocks and the accumulation of organizational capital. Matching the firm age distribution provides an additional check on how these two forces jointly shape firm survival over the life cycle.

The right panel shows the distribution of entrepreneurs by the number of years they have operated their current business. The model captures the declining pattern observed in the data. In the model, the length of time an entrepreneur operates a business is affected by both firm exit and incorporation decisions. Matching this profile provides an additional check on these decisions and their implications for entrepreneurial business tenure.

Finally, Table 5 compares the average income tax rates across top income groups in the model and the data. The data are taken from Bruggemann (2021). Although I only target the aggregate income tax revenue to GDP ratio in the calibration, the model also closely matches the average tax rates faced by different high income groups.

Table 5. Average Tax Rates for Top Percentiles of the Income Distribution

	Top 1%	Top 3%	Top 5%	Top 10%
Data	0.23	0.22	0.21	0.19
Model	0.24	0.23	0.22	0.20

5. Features of the Baseline Economy

Before turning to the policy experiments, I examine the features of the baseline economy by entrepreneurial ability. Table 6 summarizes three aspects: who becomes an entrepreneur, who incorporates, and who reaches the top income bracket.

Table 6. Features of the Baseline Economy by Entrepreneurial Ability

	Lifelong Entrepreneurs					Serial Entrepreneurs				
Ability Level	z_1	z_2	z_3	z_4	z_5	z_6	z_7	z_8	z_9	z_{10}
A. Who is an entrepreneur										
(1) Share of population (%)	40.1	38.4	13.8	5.0	1.8	0.6	0.2	0.08	0.03	0.02
(2) Share of entrepreneurs (%)	26.0	34.9	16.9	7.8	7.4	3.9	1.8	0.7	0.3	0.2
(3) Entrepreneurship rate (%)	5.8	8.2	11.0	14.2	37.2	54.0	70.2	80.5	85.9	91.6
B. Who incorporates, among those who start a business										
(4) Pr(ever incorporate) (%)	4.1	15.7	22.1	35.3	50.7	64.7	74.8	81.5	86.5	92.6
(5) Number of incorporations	0.04	0.2	0.3	0.6	1.3	2.1	3.1	4.4	5.8	7.8
(6) Median years to incorporation	32	22	18	17	14	12	9	7	6	4
C. Who reaches the top tax bracket among entrepreneurs										
(7) Share of top-bracket entrepreneurs (%)	20.1	26.4	15.1	9.0	10.1	8.4	5.8	2.9	1.3	1.0
(8) Mean income ($\times y_H$)	1.7	1.7	1.8	2.0	2.2	2.5	2.9	3.3	4.0	5.6
(9) Pr(ever reach y_H) (%)	24.5	27.4	32.0	38.6	46.5	58.3	67.9	75.9	80.2	85.8
(10) Median years to y_H	12	11	10	9	11	9	7	5	4	3

Notes: Columns correspond to the ten entrepreneurial ability levels. Row (1) reports the ability distribution among working age households, and row (2) reports the distribution among entrepreneurs. Row (3) is the share of households in each ability group who are running a business. Rows (4) and (5) report among entrepreneurs, the probability of incorporating at least once before retirement and the average number of incorporations over their working lives. Rows (6), (9) and (10) are computed at the business level. Row (6) reports the median years from founding to incorporation, conditional on incorporation. Row (7) is the ability distribution of entrepreneurs who are in the top income bracket in a given year. Row (8) reports their mean taxable income in multiples of y_H. Row (9) reports the probability that a founder's taxable income reaches y_H while operating the business. Row (10) reports the median years from founding to first reaching y_H, conditional on reaching it.

Who is an entrepreneur? Panel A of Table 6 shows the distribution of entrepreneurial ability among working age households and entrepreneurs, as well as the entrepreneurship rate within each ability group. Entrepreneurial ability is highly concentrated at the lower end of the distribution, with z_1 and z_2 accounting for 78.5% of the working age population. The entrepreneurship rate increases sharply with ability, from 5.8% for z_1 households to 91.6% for z_{10} households. Nevertheless, because most households have low entrepreneurial ability, z_1 and z_2 still account for 61% of all entrepreneurs, as shown in Row (2).

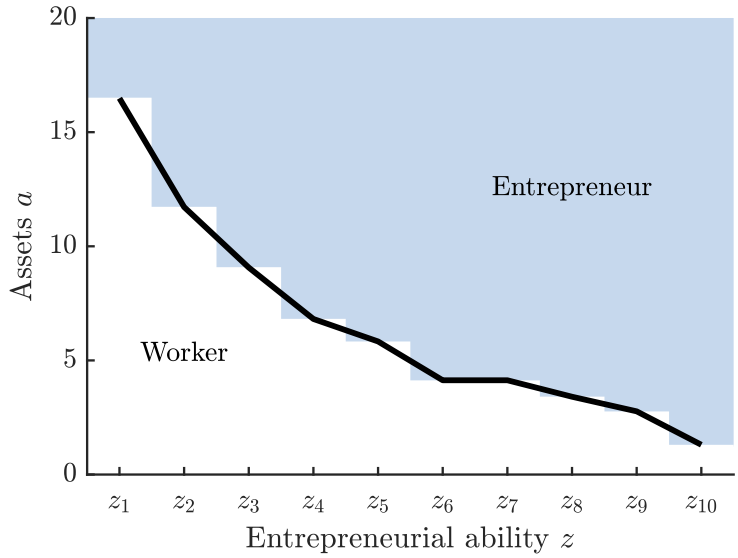


Figure 4. Entrepreneurship Entry Decision

Notes: The figure shows a worker's occupational choice as a function of current assets a and entrepreneurial ability z, with working productivity fixed at s_3, the middle of the five regular productivity states. The shaded region shows the states in which the worker chooses to start a business next period.

Figure 4 shows the entrepreneurship entry decision, which depends on three state variables: asset holdings a, working productivity s, and entrepreneurial ability z. I fix working productivity at s_3 to examine how assets and entrepreneurial ability affect entry. The minimum wealth required to start a business declines sharply with entrepreneurial ability. For z_1 households, the wealth threshold is around 17, almost four times the average wealth in the economy (4.8), whereas for z_{10} households, it is only 1.5. This difference reflects the role of wealth and entrepreneurial ability in the entry decision. Low z households have limited growth potential, but those with sufficient wealth can relax their collateral constraint and earn high enough current profits to make entrepreneurship more attractive than wage work. In contrast, high z households are willing to enter even with little wealth. Although the initial scale may be limited by their low asset holdings, their high ability allows them to accumulate organizational capital quickly, making entrepreneurship attractive through higher expected future returns.

Who incorporates? Panel B of Table 6 studies entrepreneurs' incorporation decision. The probability of incorporating increases significantly with entrepreneurial ability, from only 4.1% for z_1 entrepreneurs to 92.6% for z_{10} entrepreneurs. High ability entrepreneurs also incorporate more frequently. While entrepreneurs with ability levels z_1–z_4 incorporate less than once on average over their working lives, z_{10} entrepreneurs incorporate an average of 7.8 times.

High ability entrepreneurs also incorporate much earlier. Row (6) reports the median number of years from starting a business to incorporation, conditional on incorporation. For z_10 entrepreneurs, the median is only 4 years, compared with 32 years for z_1 entrepreneurs. This difference reflects how entrepreneurial ability affects the accumulation of organizational capital. High z entrepreneurs can accumulate organizational capital quickly, allowing them to grow their firms large enough to incorporate within a few years. In contrast, low z entrepreneurs accumulate organizational capital more slowly. Those who eventually incorporate typically need to operate their businesses for many years before doing so. These results also reveal another important feature of the model: serial entrepreneurship. High z entrepreneurs repeatedly start new businesses and incorporate them after relatively short periods. Even though each new business starts with the same initial organizational capital h_0, high z entrepreneurs can quickly build it up again and repeat the process due to their high ability.

Who reaches the top income bracket? Panel C of Table 6 turns to the top income bracket. Row (7) shows the ability distribution of entrepreneurs who are in the top income group. Because most entrepreneurs have low ability, those with ability levels z_1–z_5 still account for 80% of top-income entrepreneurs. High ability entrepreneurs are nevertheless overrepresented. Entrepreneurs with ability levels z_6–z_{10} account for less than 7% of all entrepreneurs but 20% of those in the top income bracket. They also earn much more once they are there. Row (8) shows that a z_{10} entrepreneur in the top bracket earns on average 5.6 times y_H, compared with 1.7 times for a z_1 entrepreneur.

Row (9) shows why high ability entrepreneurs are overrepresented in the top income group. The probability that a new business takes its founder to the top bracket rises from 24.5% for z_1 to 85.8% for z_{10}. Moreover, high ability entrepreneurs reach the top bracket much sooner. Among those who reach it, the median time from founding is only 3 years for z_{10} entrepreneurs, compared with 12 years for z_1 entrepreneurs.

Taken together, these results reveal two distinct entrepreneurial career paths. Households with lower entrepreneurial ability are less likely to become entrepreneurs. Those who eventually reach the top income bracket typically do so after operating the same firm for many years, gradually accumulating organizational capital along the way. Incorporation is uncommon among these entrepreneurs. I refer to entrepreneurs with ability levels z_1–z_5 as *lifelong entrepreneurs*. In contrast, households with higher entrepreneurial ability are more likely to become entrepreneurs and grow their firms rapidly. They reach the top income bracket quickly and incorporate their firms shortly thereafter. Many then start new businesses and repeat this process. I refer to entrepreneurs with ability levels z_6–z_{10} as *serial entrepreneurs*.[9]

[9]The distinction between the two entrepreneurial career paths is most pronounced at the extremes of the ability distribution. I use z_5 as the cutoff because it lies near the transition between the two paths. At this ability level, entrepreneurs have approximately a 50% probability of incorporating at least once, with an average of 1.3 incorporations over their working lives. Incorporation becomes more common and more frequent at higher ability levels.

6. Policy Analysis

In this section, I examine how an increase in the top marginal income tax rate affects the aggregate economy, with particular attention to the role of entrepreneurial decisions. In the policy experiments, I raise τ_H above its benchmark value while keeping all other model parameters unchanged. In particular, the top income threshold y_H, the other tax parameters, government spending G, and the retirement benefit $\overline{\tau}$ remain at their benchmark values. The government budget is balanced through a uniform lump-sum transfer Υ to all households, which becomes a lump-sum tax when negative. For each tax rate, I solve for a new stationary general equilibrium, allowing wages and the interest rate to adjust. The analysis focuses on steady-state comparisons without considering transition dynamics.

6.1. Laffer Curve

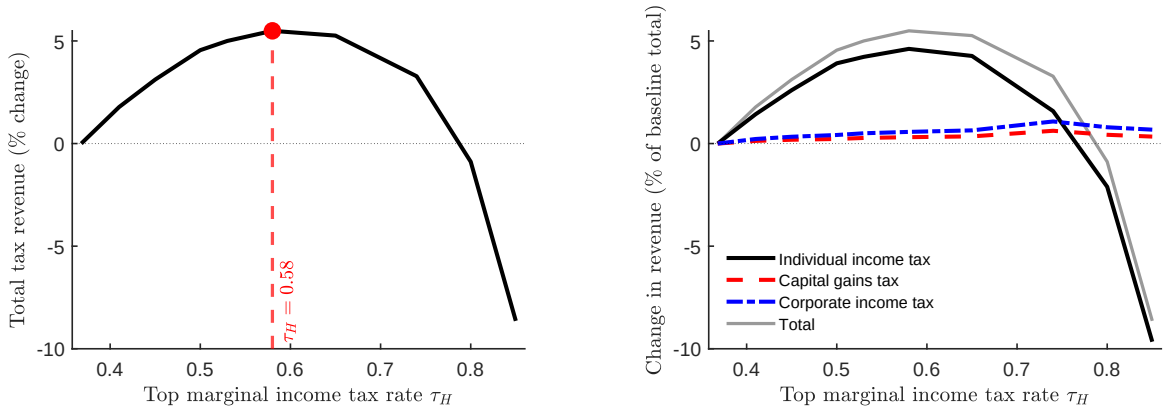
Figure 5 shows how tax revenue changes as τ_H rises from its benchmark value to 0.85. The left panel shows that total tax revenue initially increases, reaches its maximum at $\tau_H = 0.58$, and then declines. At this revenue-maximizing rate, total tax revenue is 5.5% higher than in the benchmark economy. Beyond this point, the erosion of the tax base more than offsets the revenue gains from a higher marginal tax rate.

The right panel decomposes the change in total tax revenue into personal income, corporate income, and capital gains taxes. At $\tau_H = 0.58$, personal income tax contributes 4.6 percentage points to the increase in total revenue, while corporate income and capital gains taxes contribute 0.6 and 0.3 percentage points, respectively. Personal income tax revenue drives the shape of the Laffer curve, reaching its maximum at approximately $\tau_H = 0.55$. In contrast, corporate income and capital gains tax revenues increase as τ_H rises. As entrepreneurs incorporate more firms, some income shifts from the personal income tax base to the corporate tax base, while incorporation generates capital gains. The additional revenue from these two taxes partly offsets the decline in personal income tax revenue, allowing total tax revenue to peak at a higher top marginal tax rate.

6.2. Effects on Aggregates

Table 7 reports the aggregate effects of raising the top marginal income tax rate to its revenue-maximizing level of 0.58. Compared with the benchmark economy, total output falls by 2.9%, organizational capital by 4.9%, physical capital by 6.1%, and labor by 1.4%. The interest rate rises by 9.9%, while the wage falls by 1.3%.

The higher top marginal income tax rate also changes the relative size of the two sectors. More entrepreneurs incorporate, raising the annual incorporation rate from 1.53% to 1.65% and increasing the number of corporate firms by 6.5%. Employment in the entrepreneurial sector falls by 8.9%, while employment in the corporate sector rises by 4.7%. Output follows a similar pattern, declining by 11.3% in the entrepreneurial sector and increasing by 3.3% in the corporate sector. These changes indicate a shift in production and employment toward the corporate sector.



(1) Total Tax Revenue (2) Decomposition

Figure 5. The Laffer Curve and Revenue Decomposition

Notes: The figure shows changes in tax revenue as the top marginal income tax rate increases, holding the top income threshold y_H at its benchmark level. The left panel reports the percentage change in total tax revenue relative to the benchmark economy. The right panel reports the contribution of each tax to this change, expressed as a percentage of benchmark total tax revenue. The three contributions sum to the change in total revenue (gray line).

Table 7. Aggregate and Sectoral Effects of Higher Top Income Taxation

Variable	Total	Entrepreneurial	Corporate
Aggregate Quantities			
Output (Y)	-2.93	-11.28	3.30
Organizational capital (H)	-4.92	-7.25	4.10
Physical capital (K)	-6.07	-14.91	-0.95
Labor (N)	-1.39	-8.87	4.67
Firm Mass and Per-Firm Outcomes			
Mass of firms	—	-1.05	6.48
Output per firm	—	-10.34	-2.99
Organizational capital per firm	—	-6.26	-2.24
Prices			
Interest rate (r)	9.92	—	—
Wage (w)	-1.31	—	—

Notes: The table reports percentage changes between the new steady state with $\tau_H = 0.58$ and the benchmark economy. The Total column reports economy-wide changes, while the Entrepreneurial and Corporate columns report changes within each sector. Per-firm outcomes are calculated as sectoral totals divided by the corresponding number of firms. Dashes indicate values not reported or not applicable.

Despite these differences between the two sectors, organizational capital per firm declines in both. In the entrepreneurial sector, aggregate organizational capital falls by 7.3%, while the number of firms decreases by only 1.1%. As a result, organizational capital per firm declines by 6.3%. In the corporate sector, aggregate organizational capital increases by 4.1%, but this increase is accompanied by an even larger increase of 6.5% in the number of firms. Consequently, organizational capital per firm also falls by 2.2%.

Organizational capital is particularly important because it represents the endogenously accumulated component of firm productivity in the model. To understand why organizational capital per firm declines in both sectors, I next examine how entrepreneurs adjust their organizational capital accumulation and incorporation decisions in response to the higher top income tax rate.

6.3. Heterogeneous Responses of Entrepreneurs

I first compare the organizational capital accumulation policies of the two economies. Using the policy functions from each economy, I calculate the average growth rate of organizational capital over the same benchmark distribution of entrepreneurial firms. This allows me to compare how entrepreneurs adjust their accumulation decisions while holding the distribution of firm states fixed. Panel A of Table 8 shows that entrepreneurs' accumulation responses vary across ability groups. Lifelong entrepreneurs slow down their organizational capital accumulation, with their average accumulation rate declining by 0.13 percentage points. In contrast, serial entrepreneurs accelerate their accumulation, with their average accumulation rate increasing by 0.58 percentage points. These differences are more pronounced at the extremes of the ability distribution. The accumulation rate falls by 0.22 percentage points for the bottom group (z_1–z_2), while it increases by 0.94 percentage points for the top group (z_9–z_{10}).

Table 8. Organizational Capital: Behavioral Responses and Steady-State Outcomes

	Lifelong	Serial	Bottom	Top
Ability level	z_1–z_5	z_6–z_{10}	z_1–z_2	z_9–z_{10}
A. Behavioral Responses at the Benchmark Distribution				
Change in accumulation rate (p.p.)	-0.13	0.58	-0.22	0.94
B. Steady-State Outcomes				
Change in h per firm (%)	-4.87	-12.79	-7.43	-28.22

Notes: The accumulation rate is defined as $(h' - h)/h$. Panel A evaluates the organizational capital accumulation policies of both economies on the benchmark distribution of entrepreneurial firms and reports changes in the average accumulation rate in percentage points. Panel B reports changes in average organizational capital per firm, comparing each economy using its own stationary distribution. The bottom and top groups are subsets of the lifelong and serial entrepreneurs, respectively.

Panel B compares average organizational capital per firm across the two steady states, using each economy's own distribution of entrepreneurial firms. For lifelong entrepreneurs, average organizational capital falls by 4.9%, consistent with their slower accumulation. For serial entrepreneurs, however, average organizational capital falls even more, by 12.8%, despite their faster accumulation. The contrast is most pronounced for the top group, whose average organizational capital declines by 28.2%, even though they experience the largest increase in their accumulation rate.

The results in Table 8 raise two questions. Why do lifelong and serial entrepreneurs adjust their organizational capital accumulation in opposite directions? And why do serial entrepreneurs accumulate organizational capital faster, yet operate firms with less organizational capital on average in the new steady state? The answers lie in the distinct career paths of these two types of entrepreneurs and the different roles incorporation plays in their decisions. I explain these responses through two mechanisms: the *productivity accumulation effect*, which primarily acts on lifelong entrepreneurs, and the *incorporation timing effect*, which primarily operates through serial entrepreneurs.

6.3.1. Productivity Accumulation Effect

As shown in Table 6, lifelong entrepreneurs who eventually reach the top income bracket typically do so by operating the same firm for many years and gradually accumulating organizational capital. Because of their low ability to accumulate organizational capital, rebuilding it after incorporation would take considerable time. Incorporation is less attractive to these entrepreneurs, who tend to continue operating their existing firms.

A higher top marginal income tax rate reduces the expected after-tax returns to accumulating organizational capital. Because households are forward-looking, lifelong entrepreneurs may slow down accumulation even before they reach the top income bracket. They anticipate that continued accumulation will expose more of their future income to the higher tax rate. They therefore have an incentive to accumulate more slowly and delay entering the top bracket. This dynamic response is the productivity accumulation effect. This slower accumulation reduces organizational capital per firm among lifelong entrepreneurs, contributing to the decline in the entrepreneurial sector's aggregate organizational capital.

6.3.2. Incorporation Timing Effect

Serial entrepreneurs face a different trade-off. Their high ability allows them to reach the top income bracket soon after starting a firm. Even with slower firm growth, they would soon face the higher tax rate. Slowing down organizational capital accumulation is therefore not an effective response for them. Instead, they can use incorporation to reduce their exposure to the higher personal income tax rate. By incorporating, they sell their firms and realize their value as capital gains, which are taxed at a lower rate. Their ability to build firms quickly also allows them to start new businesses and repeat this process. As the top marginal income tax rate rises, they have a greater incentive to accumulate organizational capital faster so that they can incorporate earlier and avoid the higher personal income tax.

Table 9. Incorporation and Corporate Firms by Entrepreneurial Ability

Ability level	Lifelong z_1–z_5	Serial z_6–z_{10}	Bottom z_1–z_2	Top z_9–z_{10}
A. Timing of Incorporation				
Mean firm age at incorporation, benchmark	18.11	8.24	23.47	3.88
Change in mean firm age (years)	−0.41	−0.17	−1.60	−0.14
B. Organizational Capital at Incorporation				
Mean h, benchmark	29.65	83.58	20.58	161.70
Change (%)	−2.33	−3.72	−4.43	−5.25
C. Corporate Sector				
Change in number of corporate firms (%)	8.75	1.64	17.66	2.29

Notes: Panels A and B report firm age and organizational capital at incorporation, conditional on incorporation. For each group, benchmark values and changes under $\tau_H = 0.58$ are reported. Changes in organizational capital are expressed as percentages of benchmark values, while changes in firm age are measured in years. Panel C reports the percentage change in the number of corporate firms originally founded by entrepreneurs in each ability group. The total number of corporate firms increases by 6.48%.

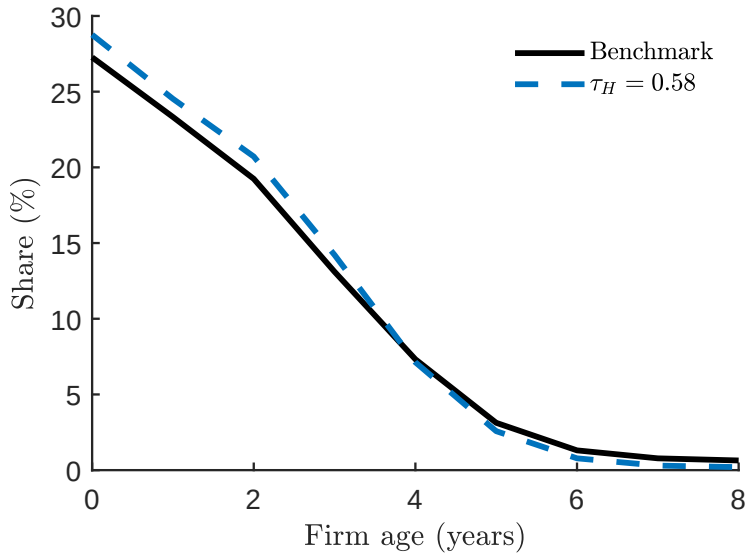


Figure 6. Firm Age Distribution for the Top Ability Group

Notes: The figure compares the age distribution of entrepreneurial firms operated by the top ability group in the benchmark economy and the new steady state with $\tau_H = 0.58$. The distributions include only firms that remain in the entrepreneurial sector. Firm age is measured in years since founding, starting at zero.

This is confirmed by Panel A of Table 9, which shows that the average firm age at incorporation falls by 0.17 years for serial entrepreneurs. They also incorporate with less organizational capital. Panel B shows that average organizational capital at incorporation falls by 3.7% for serial entrepreneurs and by 5.3% for the top group. These results are consistent with serial entrepreneurs using incorporation as a tax-advantaged exit. They accumulate organizational capital faster to incorporate sooner, selling their firms at a younger age and with less organizational capital.

Earlier incorporation also changes the composition of firms in the entrepreneurial sector. Serial entrepreneurs sell their firms sooner and at lower levels of organizational capital, leaving fewer mature firms with high h in the entrepreneurial sector. Many then start new businesses with initial organizational capital h_0. As a result, a larger share of their firms are young and have accumulated relatively little organizational capital. Figure 6 shows this shift for the top ability group. The average age of their entrepreneurial firms falls from 2.2 to 1.6 years, while the share of firms in their first three years increases. This explains why serial entrepreneurs operate firms with less organizational capital on average, despite accumulating it faster. Earlier incorporation then reduces the organizational capital held in the entrepreneurial sector.

6.4. Incorporation and the Reallocation of Organizational Capital

I now turn to organizational capital in the corporate sector. As shown in Table 7, aggregate organizational capital increases by 4.1%, driven by a 6.5% increase in the number of corporate firms. Average organizational capital per firm, however, falls by 2.2%. Unlike entrepreneurial firms, corporate firms are operated to maximize their present value, so their accumulation decisions are not directly affected by changes in the personal income tax rate, apart from general equilibrium effects. Hence, the most direct channel through which the tax reform affects organizational capital in the corporate sector is the inflow of firms and the organizational capital they bring with them at incorporation. I examine how this inflow changes below.

As discussed above, serial entrepreneurs incorporate earlier and with less organizational capital, bringing less mature firms into the corporate sector. This lowers average organizational capital per firm in the corporate sector. More importantly, the tax reform also changes the incorporation decisions of some lifelong entrepreneurs. As shown in Table 6, lifelong entrepreneurs who eventually reach the top income bracket typically operate the same firm for many years. Even after reaching the top bracket, they continue operating their firms and accumulating organizational capital for many more years before incorporating. For example, among z_1 entrepreneurs, the median time from starting a firm to reaching the top income bracket is 12 years, and they typically continue operating the same firm for another 20 years before incorporating. Given their low ability, selling an established firm means giving up a business that would take many years to rebuild. They therefore prefer to retain their firms, even after reaching the top income bracket, until the value of incorporating is sufficiently high to compensate for giving up continued ownership.

A higher top marginal income tax rate changes this trade-off. Although lifelong entrepreneurs slow down organizational capital accumulation to delay entering the top income bracket, some eventually reach it. Once they do, continuing to operate their firms exposes their business profits to the higher personal income tax rate, whereas incorporation allows them to realize their firms' value at the lower capital gains tax rate. The tax advantage of incorporation now outweighs the benefits

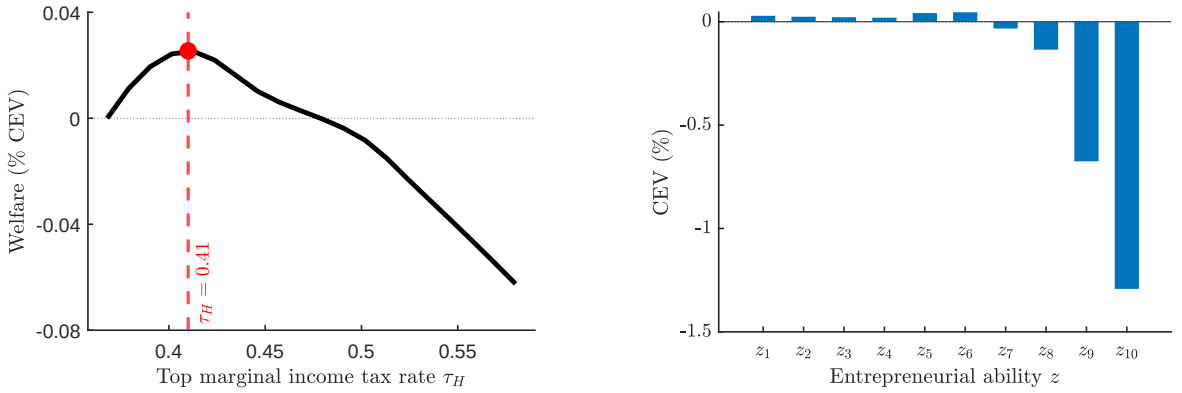
of continued ownership, inducing them to incorporate firms they would otherwise have continued to operate in the benchmark economy.

Panel C of Table 9 shows that the number of corporate firms originally founded by lifelong entrepreneurs increases by 8.8%. The increase is particularly large for the bottom ability group (z_1–z_2), at 17.7%. Firms incorporated by lifelong entrepreneurs carry substantially less organizational capital than those incorporated by serial entrepreneurs. As shown in Panel B, their average organizational capital at incorporation is 29.7, compared with 83.6 for serial entrepreneurs. The increased inflow of these relatively low h firms therefore puts further downward pressure on average organizational capital in the corporate sector. At the same time, the entrepreneurial sector loses established firms with relatively high h within the lifelong entrepreneur group, further reducing its organizational capital.

Together, earlier incorporation by serial entrepreneurs and increased incorporation by lifelong entrepreneurs reallocate organizational capital from the entrepreneurial sector to the corporate sector. The inflow of relatively low h firms reduces average organizational capital per corporate firm, even though the larger number of firms raises aggregate organizational capital in this sector. However, this increase does not offset the decline in the entrepreneurial sector, leaving the economy with less organizational capital overall.

6.5. Welfare Implications

I now turn to the welfare implications of the tax reform. Since the analysis focuses on steady states, I evaluate welfare for a newborn under the veil of ignorance using the consumption-equivalent variation (CEV). The CEV measures the uniform percentage change in consumption across all periods and states required to make a newborn in the benchmark economy indifferent to being born into the new steady state.



(a) Aggregate Welfare (b) Welfare by Entrepreneurial Ability

Figure 7. Welfare Effects of Higher Top Income Taxation

Notes: Welfare is measured by the consumption-equivalent variation (CEV) of a newborn relative to the benchmark steady state. Panel (a) shows ex ante welfare under the veil of ignorance as the top marginal income tax rate τ_H increases. The figure is truncated at the revenue-maximizing rate of 0.58 for readability. Panel (b) reports welfare conditional on entrepreneurial ability at the welfare-maximizing rate, $\tau_H = 0.41$.

The left panel of Figure 7 shows how welfare changes as the top marginal income tax rate increases. Welfare initially rises, reaching its maximum at $\tau_H = 0.41$, with a CEV of only 0.03%. This is well below the revenue-maximizing rate of 0.58. Welfare turns negative at around $\tau_H = 0.48$ and continues to decline as the tax rate rises.[10]

The gap between the welfare-maximizing and revenue-maximizing tax rates reflects the diminishing revenue gains from higher taxation relative to its welfare costs. A higher top marginal income tax rate generates revenue to finance larger lump-sum transfers, but it also imposes welfare costs. As discussed above, higher top tax rates induce lifelong entrepreneurs to slow down organizational capital accumulation and serial entrepreneurs to incorporate earlier, reducing organizational capital and aggregate output. These responses also erode the personal income tax base. As the Laffer curve shows, total tax revenue continues to rise until $\tau_H = 0.58$, but the curve is relatively flat near its peak. Further increases in the tax rate generate only modest additional revenue, while the welfare costs from distorted entrepreneurial decisions and higher taxes on top incomes continue to rise. Beyond $\tau_H = 0.41$, the welfare gain from the additional transfer is no longer sufficient to offset these costs.

The right panel of Figure 7 reports the welfare effects of the reform at $\tau_H = 0.41$ by entrepreneurial ability. Newborns in the six lowest ability groups (z_1–z_6) gain between 0.02% and 0.05%. In contrast, newborns in the four highest ability groups (z_7–z_{10}) experience welfare losses, which increase sharply with ability, reaching 0.68% for z_9 and 1.29% for z_{10}. Since the highest ability groups account for only a small fraction of the population, the modest gains of the majority outweigh their larger losses, resulting in a small positive aggregate welfare gain.

Low-ability households receive a modest increase in the lump-sum transfer, but some also face higher income taxes. Although most remain below the top income bracket, others may enter it either by receiving a superstar labor productivity shock as workers or by operating the same firm for many years as entrepreneurs. Hence, low-z households only have small expected welfare gains.

The highest ability households, by contrast, experience substantial welfare losses because they are much more likely to become top income earners. They typically follow the serial entrepreneur career path, reaching the top income bracket shortly after starting a firm and earning three to six times the top income threshold y_H. Consequently, the higher tax rate affects a larger fraction of households in these ability groups and applies to a larger share of their income.

7. Conclusion

This paper studies the aggregate effects of top income taxation in an economy where entrepreneurs accumulate organizational capital over the firm life cycle and can transfer their businesses to the corporate sector through incorporation. The central finding is that higher top marginal income tax rates affect both how entrepreneurs build their firms and when they incorporate, with responses differing substantially by entrepreneurial ability. These decisions reshape firm dynamics and the allocation of organizational capital between the entrepreneurial and corporate sectors,

[10] For readability, the figure shows top tax rates only up to the revenue-maximizing rate of 0.58. I also solve the model for rates up to 0.85. Welfare continues to decline beyond 0.58, reaching -3.1% at $\tau_H = 0.85$.

extending the effects of top income taxation beyond the incomes of top earners. In the calibrated economy, raising the top marginal income tax rate from 0.37 to the revenue-maximizing rate of 0.58 reduces aggregate output by 2.9% and organizational capital by 4.9%.

The analysis identifies two mechanisms. The productivity accumulation effect slows organizational capital accumulation among lifelong entrepreneurs, while the incorporation timing effect induces serial entrepreneurs to accelerate accumulation and incorporate earlier. Together, these responses reduce organizational capital in the entrepreneurial sector. Higher top income taxation also induces some lifelong entrepreneurs to incorporate firms they would otherwise have continued operating. These changes increase the number of firms in the corporate sector but reduce average organizational capital per firm. Although aggregate organizational capital rises in the corporate sector, this increase is insufficient to offset the decline in the entrepreneurial sector.

These entrepreneurial responses also shape the revenue and welfare consequences of top income taxation. Slower firm growth erodes the personal income tax base, while changes in incorporation shift tax revenue toward capital gains and corporate taxation. Although total tax revenue continues to increase up to a top marginal income tax rate of 0.58, the additional revenue becomes limited near the peak of the Laffer curve, while the welfare costs of higher taxation continue to rise. As a result, the welfare-maximizing tax rate is considerably lower, at 0.41.

The framework developed in this paper can be used to study a broader range of policies affecting entrepreneurship and firm dynamics. One natural extension is to examine the joint design of personal income taxes, capital gains taxes, and corporate income taxes. By allowing entrepreneurs to accumulate organizational capital and choose when to incorporate, the model captures how the relative taxation of business profits and incorporation shapes both firm growth and the allocation of organizational capital across sectors. While this paper holds capital gains and corporate income tax rates fixed to isolate the effects of top income taxation, the framework provides a basis for studying how these tax instruments interact and their implications for aggregate productivity and welfare.

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Appendices

A. Legal Forms of Organization in the U.S.

Business owners in the U.S. can choose among several legal forms of organization, subject to applicable state laws. The main forms are sole proprietorships, general partnerships, limited partnerships, limited liability companies (LLCs), S corporations, and C corporations. Their main characteristics are summarized in Table 10.

C corporations face no restrictions on the number of shareholders and can accommodate a broader range of owners. They generally have greater access to external equity financing. Decision-making authority rests with the board of directors rather than directly with shareholders. In the U.S., income earned by a C corporation is subject to two layers of taxation. The corporation first pays corporate income tax on its profits at the entity level. Shareholders then face a second layer of taxation, either when profits are distributed as dividends or when they realize capital gains by selling their shares.

Sole proprietorships, general partnerships, limited partnerships, LLCs, and S corporations are pass-through entities: business income passes through to the owners and is taxed at the owner level. Sole proprietorships have a single owner, whereas the other pass-through forms can have multiple owners. Owners of sole proprietorships and general partnerships generally do not receive limited-liability protection, while limited partnerships and LLCs provide partial or full protection from business liabilities.

Table 10. Main Characteristics of Different Legal Forms of Organization

	Number of owners	Taxation
Sole Proprietorship	1	Pass through
General Partnership	> 1	Pass through
Limited Partnership	≥ 1	Pass through
Limited Liability Company	≥ 1	Pass through
S Corporation	$1 - 100$	Pass through
C Corporation	Unlimited number of shareholders	Entity level

Entrepreneurial firms in the model most closely resemble sole proprietorships, as each firm has a single owner. According to the Integrated Business Data, sole proprietorships and partnerships are the most common forms of business organization in the U.S., accounting for more than 80% of all businesses. S corporations are also typically closely held. Although they can have up to 100 shareholders, about 94% have three or fewer shareholders (U.S. Government Accountability

Office, 2009). In contrast, I view the corporate sector in the model as representing firms with access to outside equity and potentially dispersed ownership. C corporations are therefore a more appropriate counterpart for this sector.

B. Welfare

This appendix explains how I compute the newborn CEV reported in Section 6.5. I index the benchmark economy by $j = 0$ and the economy with the higher top rate by $j = 1$.

Period utility is additively separable in consumption and labor supply. Hence, lifetime utility can be split into a consumption part and a labor part. For a household in state x under the decisions of economy j, define

$$A_j(x) = E\left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma_1}}{1-\sigma_1} \,\middle|\, x\right], \qquad B_j(x) = E\left[\sum_{t=0}^{\infty} \beta^t \chi \frac{l_t^{1+\sigma_2}}{1+\sigma_2} \,\middle|\, x\right], \tag{20}$$

where the expectations are taken over the rest of the household's life, which ends at death, under the optimal decisions of economy j. The value function is then $V_j(x) = A_j(x) - B_j(x)$.

When a newborn enters the economy, she inherits the assets of a retiree who has just died, and draws her working productivity from $\mathcal{F}_s$ and her entrepreneurial ability from $\mathcal{F}_z$. A retiree chooses her savings a' and then dies with probability Ω_d, so a newborn's inheritance equals the savings of a dying retiree. Because death does not depend on the retiree's state, the distribution of inheritances in economy j is the distribution of retirees' savings choices:

$$\psi_j(\mathcal{A}) = \frac{1}{\mu_{OR}} \int \mathbf{1}\{a'_j(a) \in \mathcal{A}\}\, d\Lambda_j^{OR}(a), \tag{21}$$

where Λ_j^{OR} is the stationary distribution of retirees over assets, μ_{OR} is their mass, and $a'_j(a)$ is the retiree's savings policy. Hence, newborns in economy j are distributed as

$$\Lambda_j^n(a, s, z) = \psi_j(a)\, \mathcal{F}_s(s)\, \mathcal{F}_z(z). \tag{22}$$

$\mathcal{F}_s$ and $\mathcal{F}_z$ are exogenous and the same in both economies. I write $E_j[\cdot]$ for the expectation over Λ_j^n. Newborn welfare in economy j is $W_j = E_j[V_j]$, where $V_j(a, s, z)$ is the value of a young household.

The CEV λ is the percentage increase in consumption, in every period and state, that makes a newborn in the benchmark economy as well off as a newborn in the new steady state. Hours and all other decisions are held at their benchmark values. The CEV solves

$$E_0\left[(1+\lambda)^{1-\sigma_1} A_0 - B_0\right] = E_1[V_1], \tag{23}$$

which gives

$$\lambda = \left(\frac{E_1[V_1] + E_0[B_0]}{E_0[A_0]}\right)^{\frac{1}{1-\sigma_1}} - 1. \tag{24}$$

C. Algorithm: Solving the Household and Firm Problems

This appendix summarizes how I solve the household and corporate firm problems for given prices (r, w) and tax policy. All problems are solved by value function iteration on discrete grids. To speed up convergence, I use Howard's policy improvement: after each maximization step, I hold the policy functions fixed and iterate the resulting linear Bellman equation 30 times before maximizing again.

I reduce the firm's input choice to one dimension. Capital and labor enter production through the composite input $X = k^{\alpha} n^{1-\alpha}$. For a given X, a firm chooses k and n to minimize the cost $wn + (r + \delta)k$. Without the collateral constraint, the minimized cost is μX, where $\mu = \big((r + \delta)/\alpha\big)^{\alpha}\big(w/(1-\alpha)\big)^{1-\alpha}$. If the cost-minimizing capital is greater than ηa, capital is set to ηa and labor adjusts to deliver X. Output $y = q h^{1-\gamma} X^{\gamma}$ and next-period organizational capital $h' = (1 - \delta_h)h + z y^{\psi}$ depend on the inputs only through X. The input decision then reduces to a choice of X. Given the chosen X, I recover k and n from the cost-minimization problem.

I solve the corporate firm and household problems using the following steps.

Step 1: Corporate firms. I solve for the firm value $\Pi(h, q, z)$ separately for each z. In each state (h, q), the firm chooses X. Since the implied h' generally falls between grid points, the continuation value is interpolated linearly in h.

Step 2: Retirees. The retiree problem depends only on r and the tax schedule. I solve it with savings chosen from the asset grid. Neither the retiree value nor the corporate firm value depends on the working-age value functions. I therefore solve them first and treat them as fixed in Step 3.

Step 3: Working-age households. I solve the value functions of workers, entrepreneurs, and incorporation entrepreneurs jointly, because each enters the others' continuation values through the occupation choice. Each iteration proceeds as follows.

1. I compute the expected value of each occupation next period, integrating over s' and q' with their transition matrices. For a new business, I set organizational capital to h_0, integrate over q' using $\mathcal{F}_q$, and deduct the entry cost C_e from assets.
2. **Workers.** Given (a, s, a'), labor supply is a static choice. I solve its first-order condition by bisection and store the resulting period utility for every (a, s, a') before the iteration begins. Savings are then chosen from the asset grid.
3. **Entrepreneurs.** In each state, I search jointly over X and a'. The continuation value at h' is interpolated in h.
4. **Incorporation entrepreneurs.** Their resources include the after-tax sale proceeds $(1 - \tau_k)\Pi(h, q, z)$ from Step 1, and they choose savings from the asset grid.
5. **Occupation choices.** At the chosen a' (and h' for entrepreneurs), each household picks the occupation with the highest expected value. Because the choice is made before s' and q' are realized, it compares expected values rather than realized ones.

The Howard steps then hold savings, inputs, labor, and occupation choices fixed. For each z, the problems are independent given the expected values, so I solve them in parallel across z.